Character Design Report

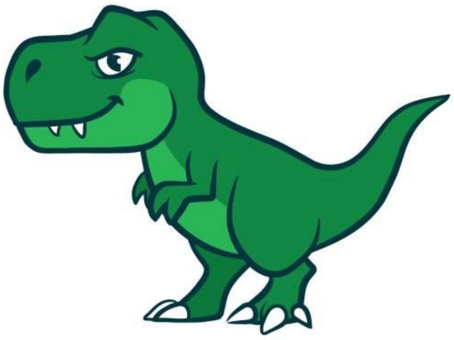
Felicia Jolie

Contents:

1 Research

References:

Reference 1: Tyrannosaurus Rex



- I find the proportions of the Tyrannosaurus rex both cool and intimidating, making it an ideal reference for my model.
- To create the body, I start with a cube, scaling and resizing it to achieve the general shape of a dinosaur.
- Additionally, I used the extrusion technique on the cube to form the arms and legs, refining the model's overall structure.

Reference 2: Stegosaurus

- I admire the spikes along the back of the Stegosaurus, as well as its tail, and I plan to incorporate similar features into my model.
- To create the spikes, I extrude sections from the back of the cube to form three spikes. I then refine their shape by extruding further and scaling them to achieve a sharp, pointy appearance.
- To create the tail, I extrude the lower section of my model's body and gradually scale it while extruding to achieve a tapered, pointy shape.

Reference 3: Mike Wazowski (Monsters, Inc.)

- I find Mike Wazowski's single eye adorable and want to incorporate a similar feature into my model.
- To create the eye, I start with a sphere, resizing and adjusting it to resemble a realistic eye.
- I then position and adjust it on the head of my model to make it look like an eye

Reference 4: Devil

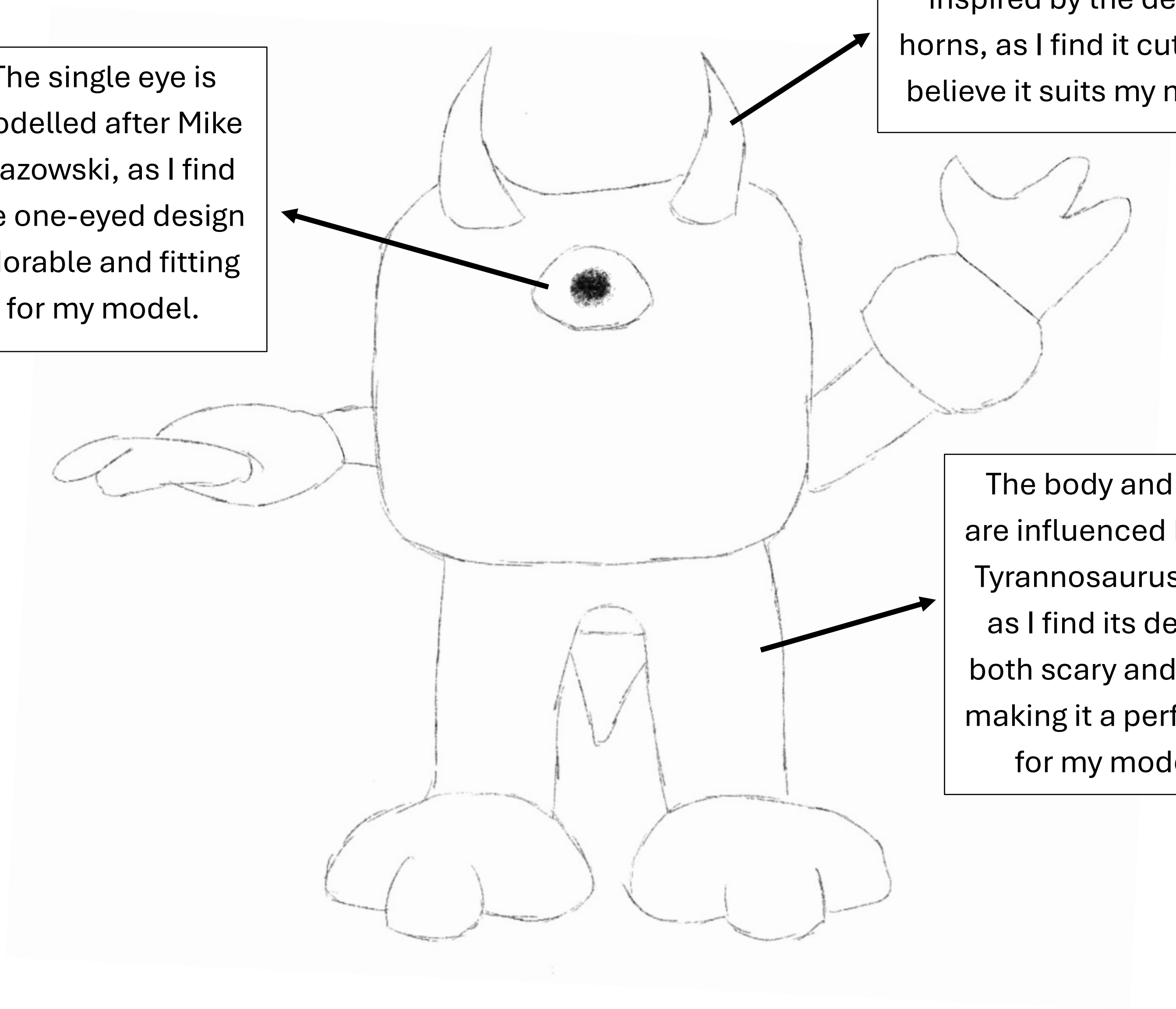
- The curved horns of the devil inspire me, and I believe they will complement my model's design.
- To create the horns, I extrude the top part of the head and scale it to form a sharp, pointy shape.
- I then carefully curve the horns to replicate the distinctive appearance of the devil's horns.

Reference 5: Alien (Toy Story)

- I like the vibrant colours of the aliens in Toy Story and feel that my model will look great in a similar palette.
- Using Adobe Substance 3D Painter, I plan to recreate their signature look by incorporating of green, light and dark blue, and purple, ensuring the colours complement and enhance the overall design of my model.

2 Design Development

Front Part:

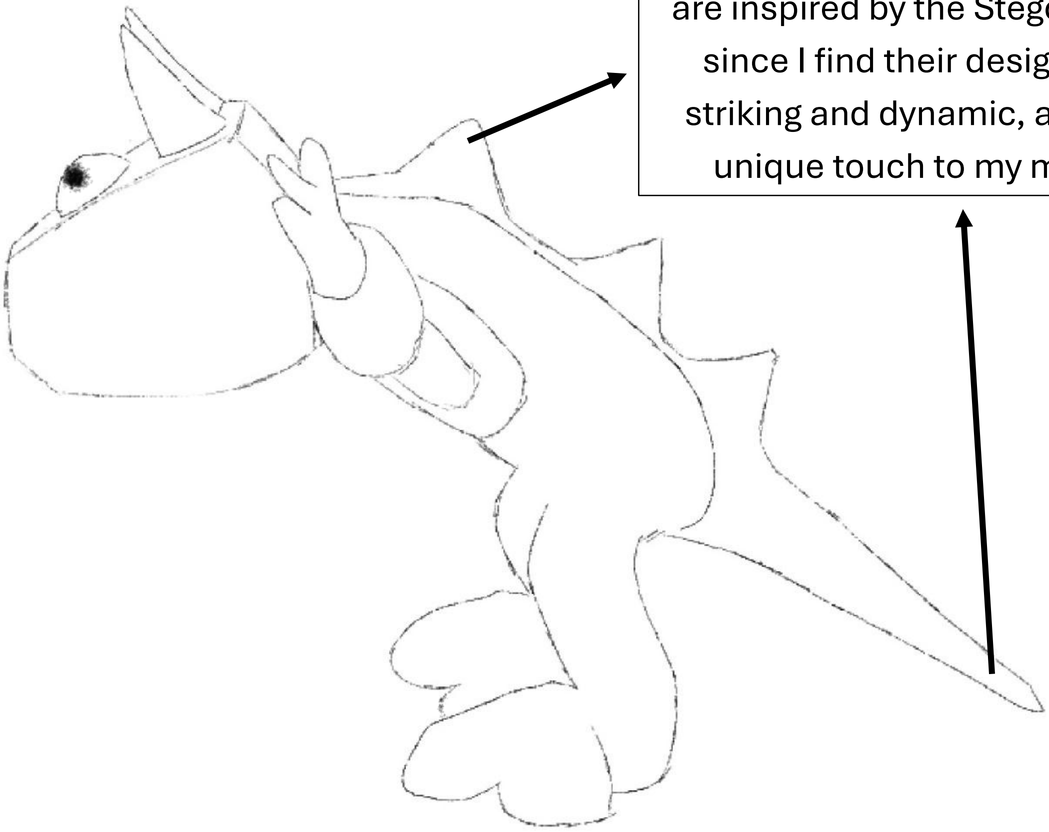


The horns design is inspired by the devil's horns, as I find it cute and believe it suits my model

The single eye is modelled after Mike Wazowski, as I find the one-eyed design adorable and fitting for my model.

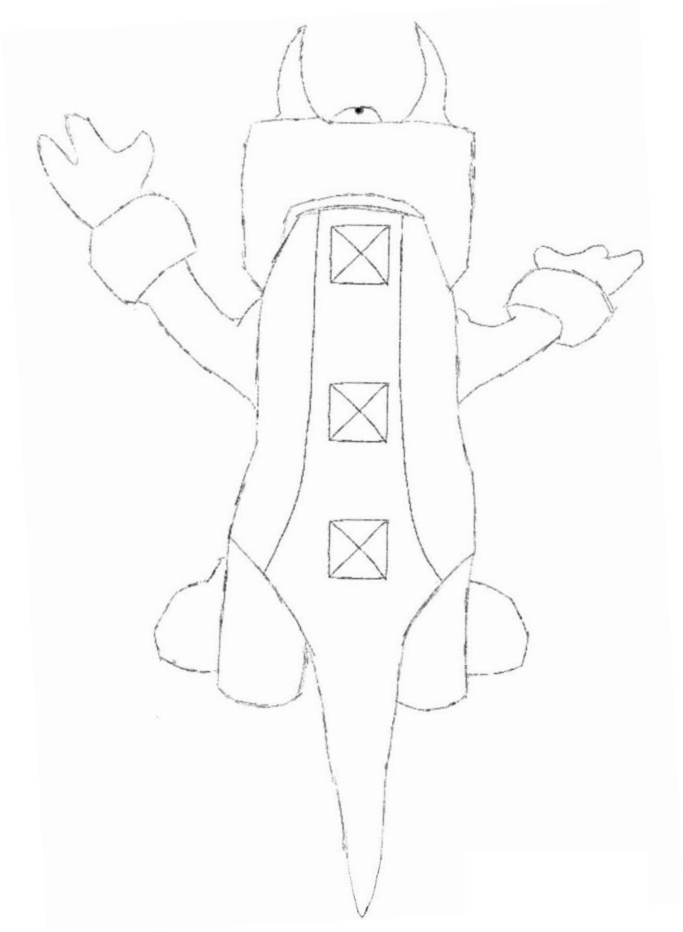
The body and legs are influenced by the Tyrannosaurus Rex, as I find its design both scary and cool, making it a perfect fit for my model.

Side Part:



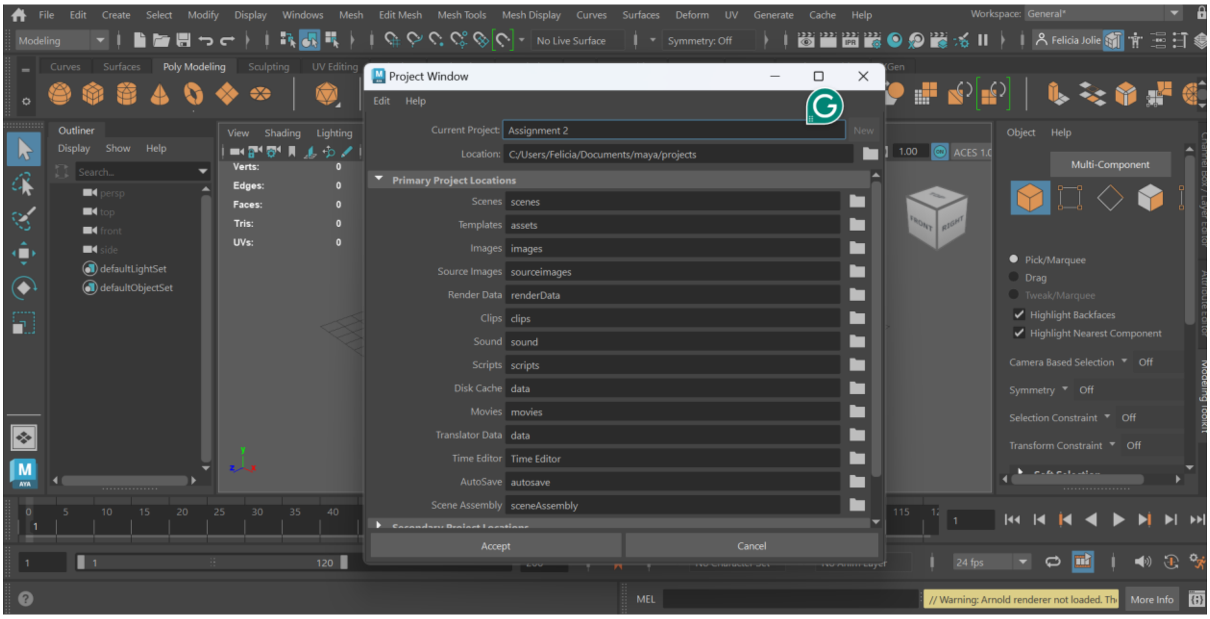
The spike on the back and the tail are inspired by the Stegosaurus, since I find their design both striking and dynamic, adding a unique touch to my model.

Top Part:

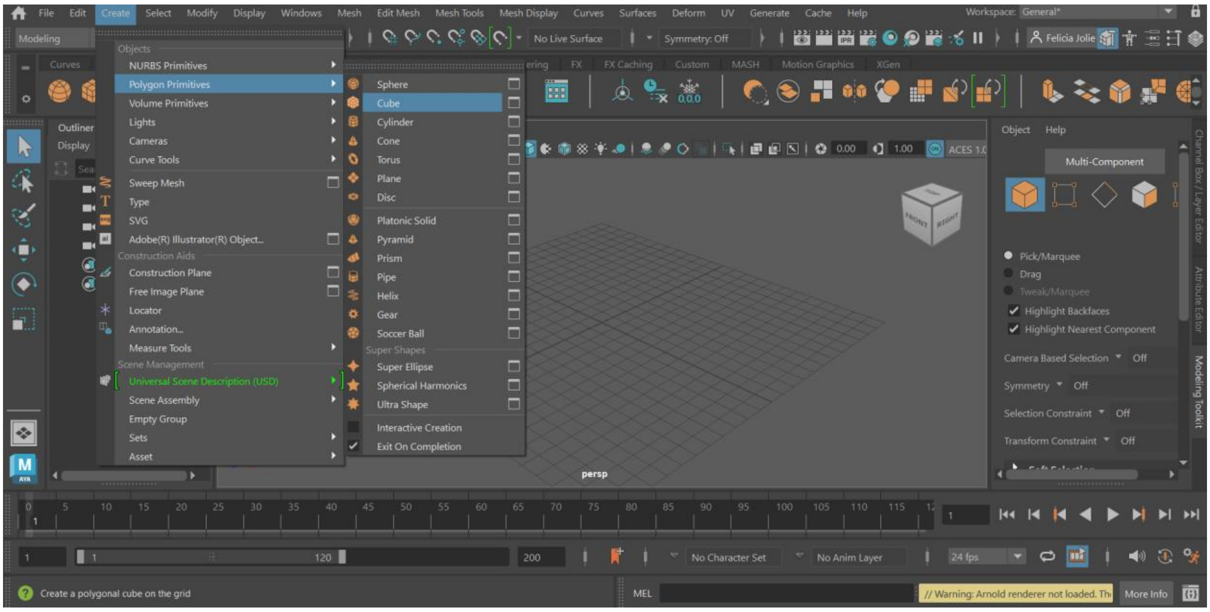


3 Maya Modelling Process

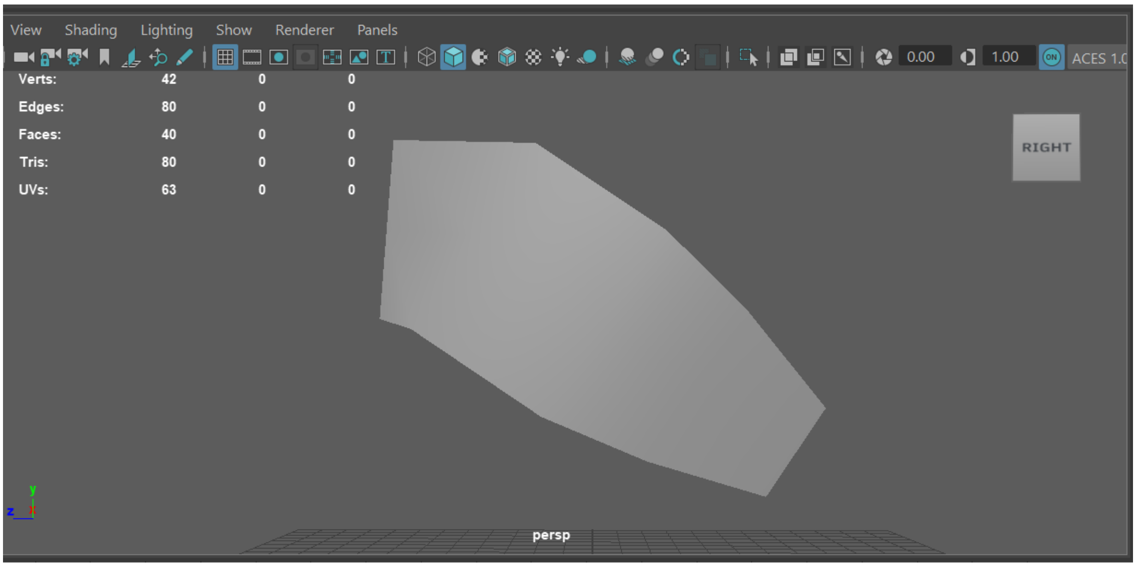
1. First, I create a new Maya project and save it as Assignment 2 in my folder.



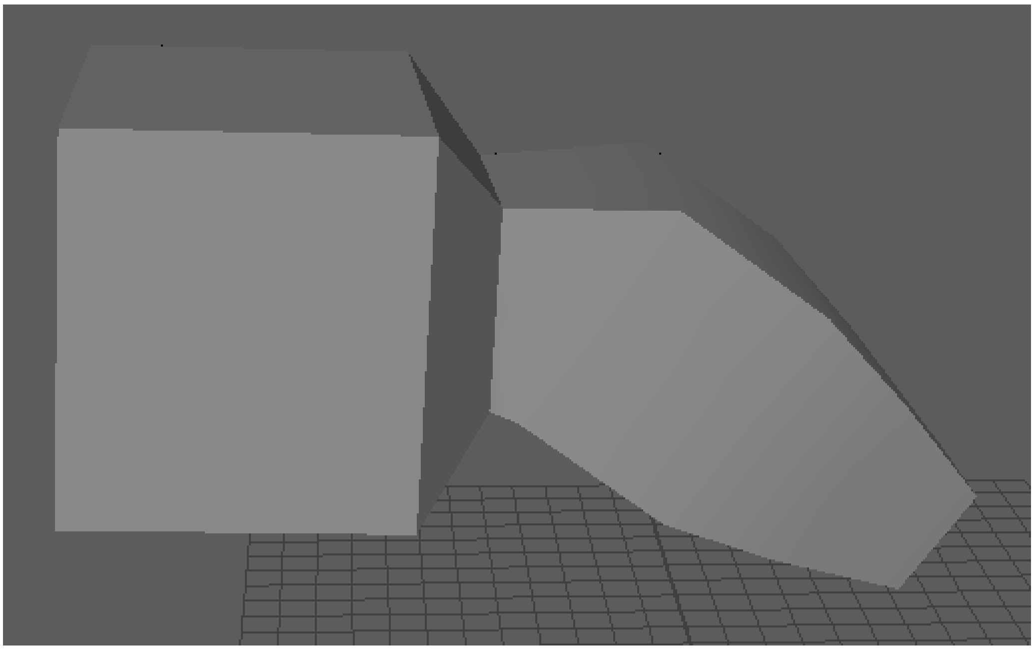
2. Then I create a cube by going to Create → Polygon Primitives → Cube.



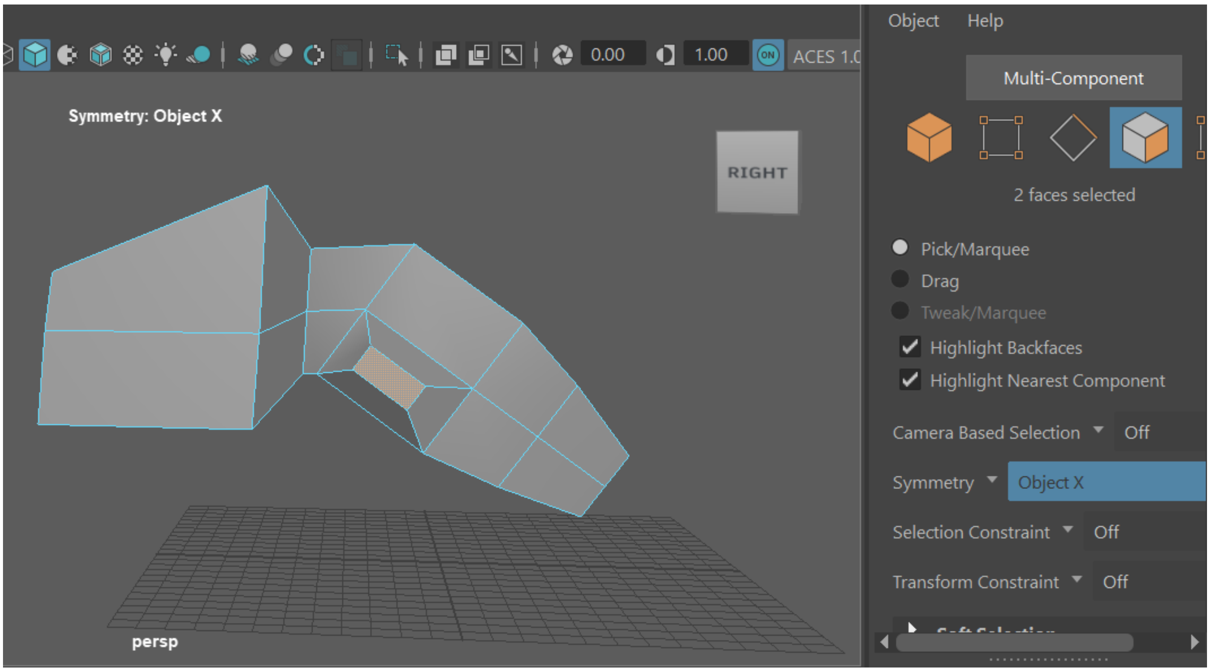
3. I scale and rotate the cube using the move, scale, and rotate tool to shape the body of my model.



4. I extrude the top part of the body to make the head of my model.

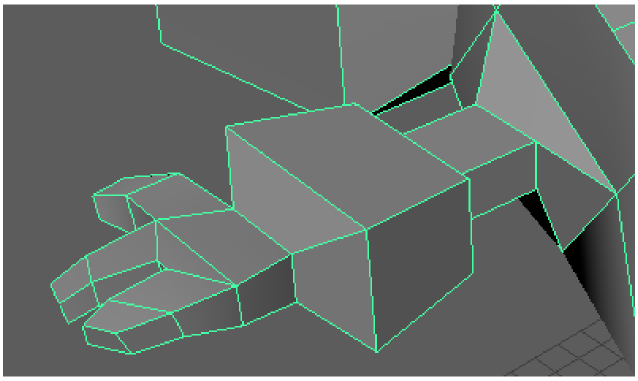


5. I turn on the Symmetry Object X then extrude the top part of the body to make both of the hands together.

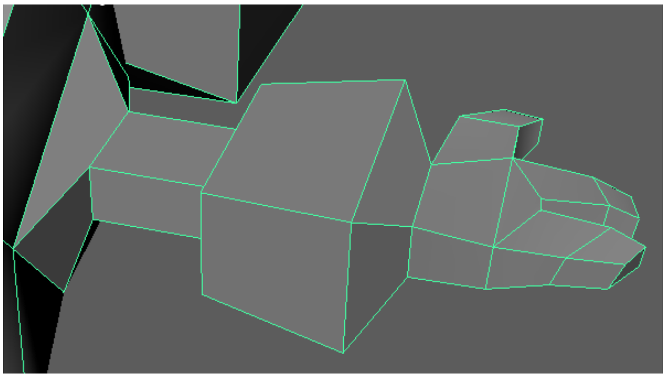


6. I extrude the hands more and shape using the move, scale, and rotate tool so it look like a hand. I make both of the hands at the same time because I turn on the Symmetry Object X.

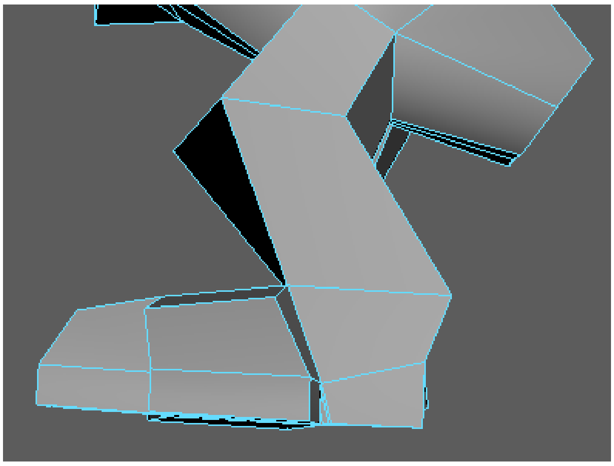
Right Hand

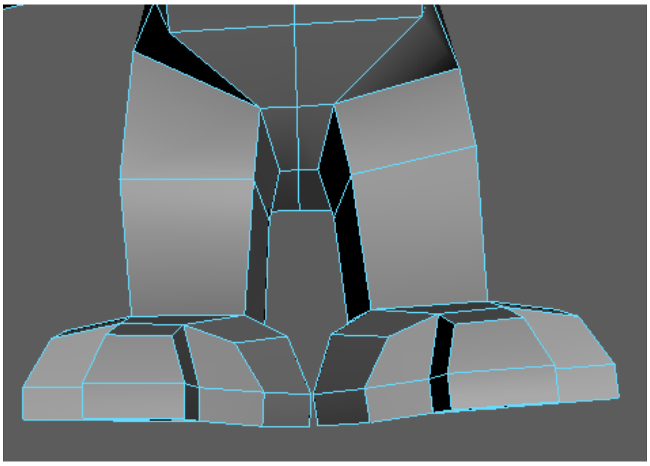


Left Hand

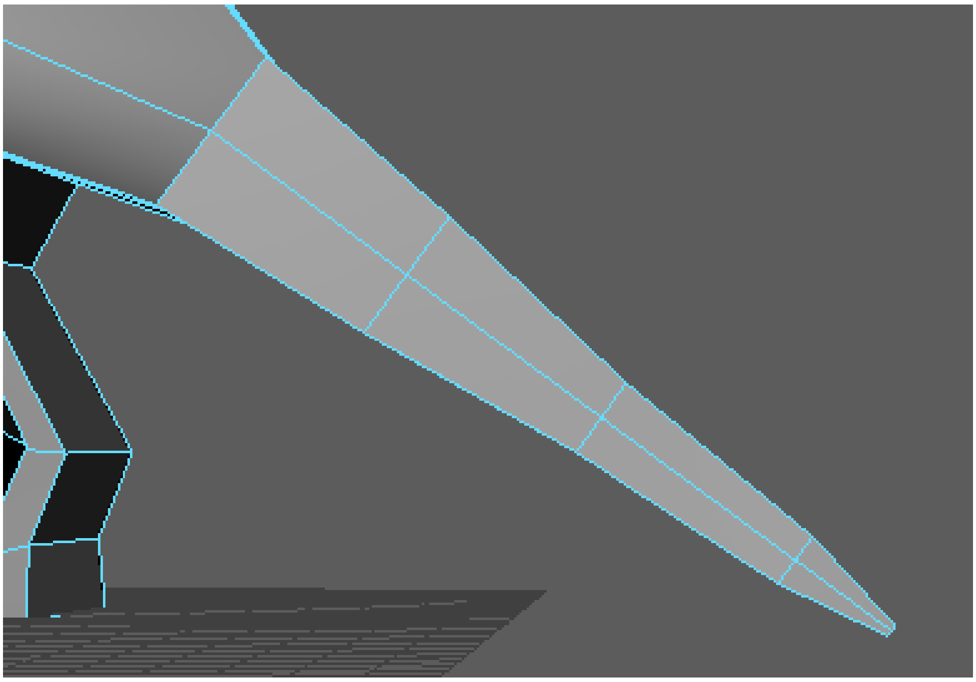


7. While the Symmetry Object X is still turned on, I extrude the bottom part of the body to make the legs and shape it using the move, scale, and rotate tool. I turned the Symmetry off then I extruded the bottom part of the body to make the tail.

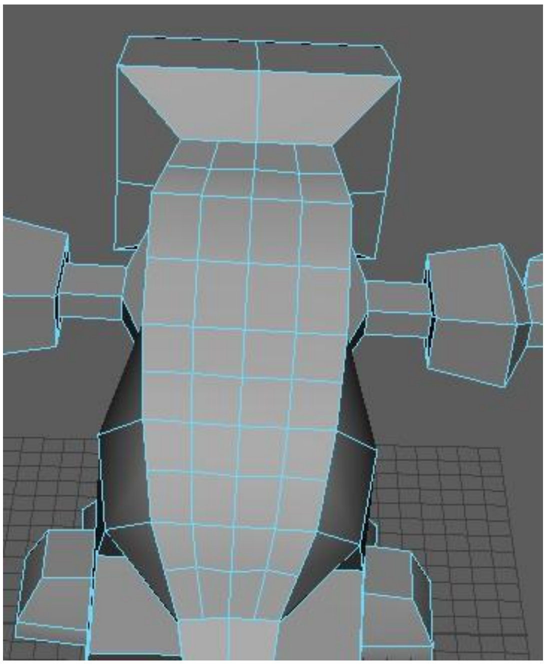




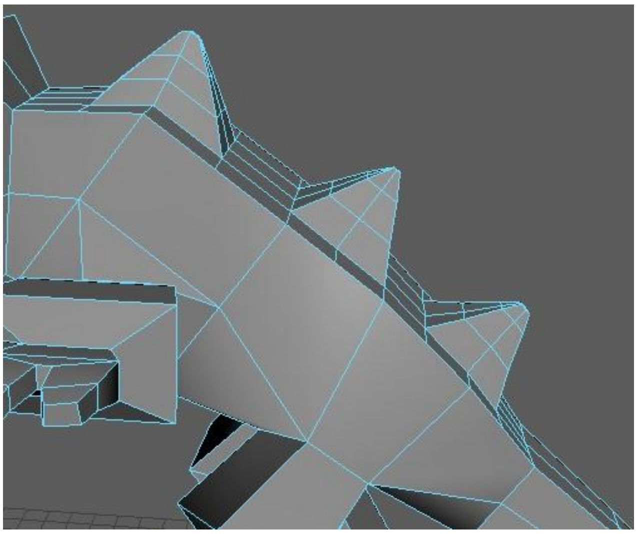
8. I extrude the bottom part of the body to make the tail. To make the tail long and pointy, I extrude and and scale it 4 times.

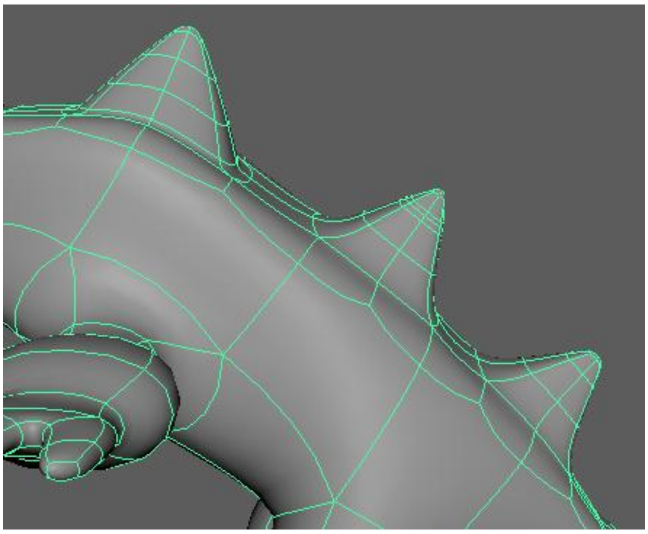


9. I make a lot of divisions at the top of the body to make the spikes at the back.

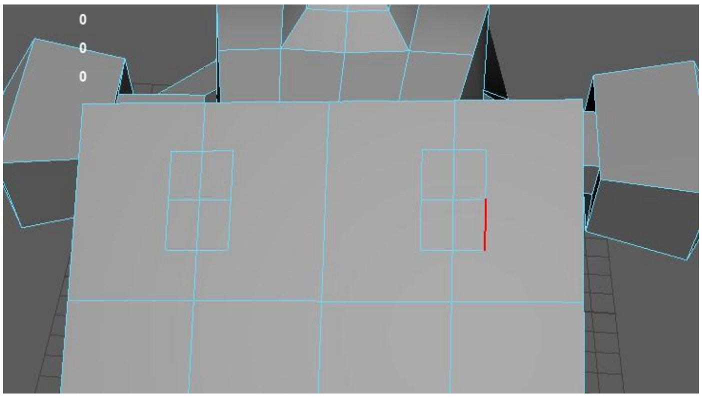


10. I make the 3 spikes for the back by using the move, scale, and rotate tool. To make the spike long and pointy, I extrude and scale it 3 times.

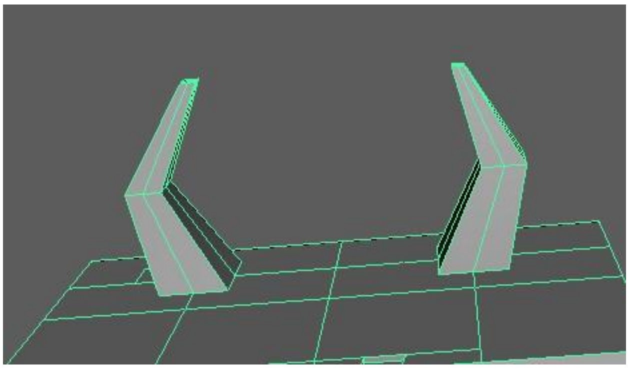


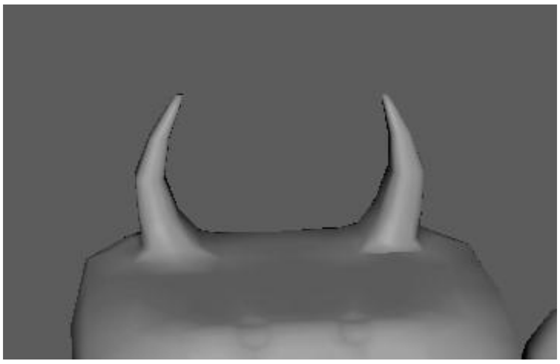


11. I make some divisions at the top of the head to make the horns.

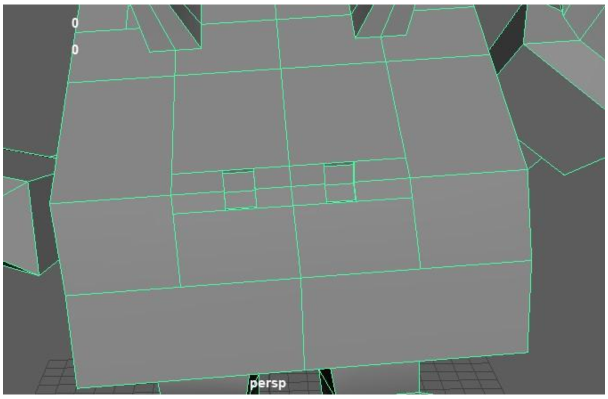


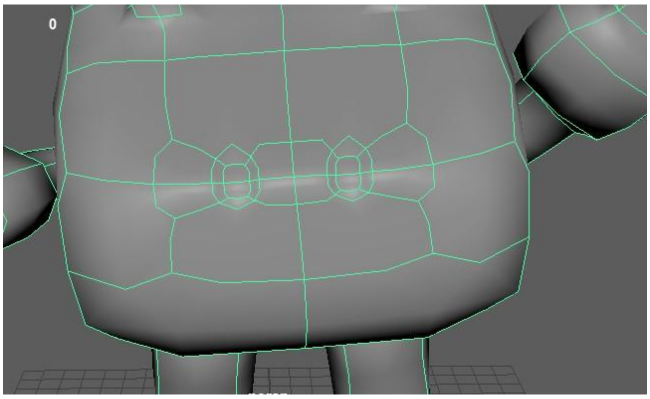
12. Then I extrude it twice and scale it using the scale tool to make it look like a horn. I then select the vertex of the middle part and move it using move tool to make it curve.



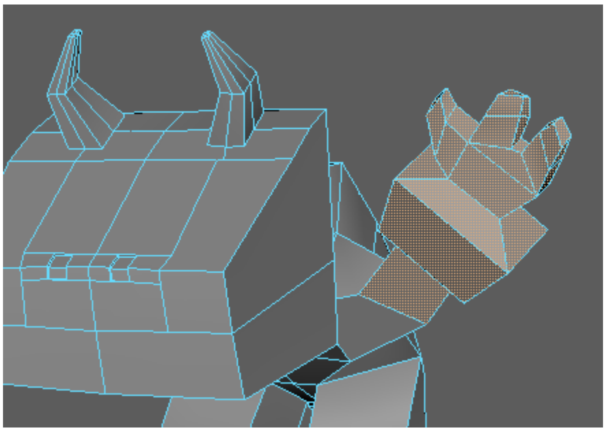
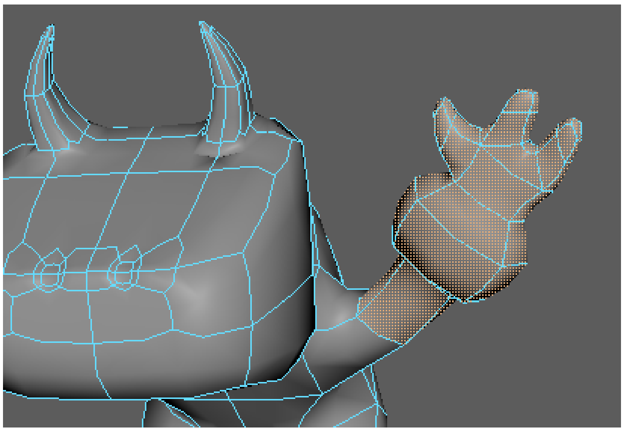


13. I make some division in the head and then extrude it inwards to make the nose of my model.

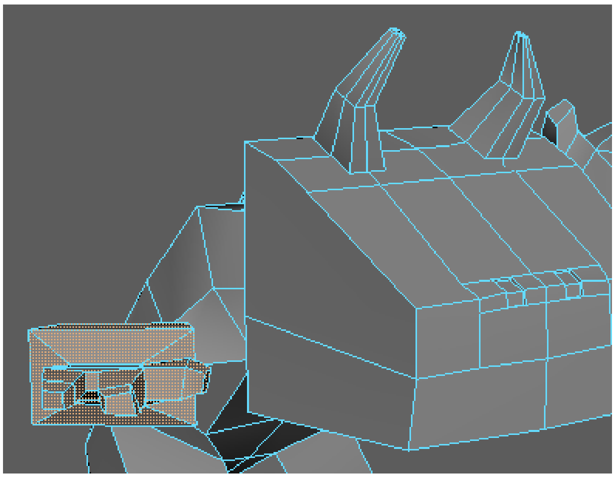
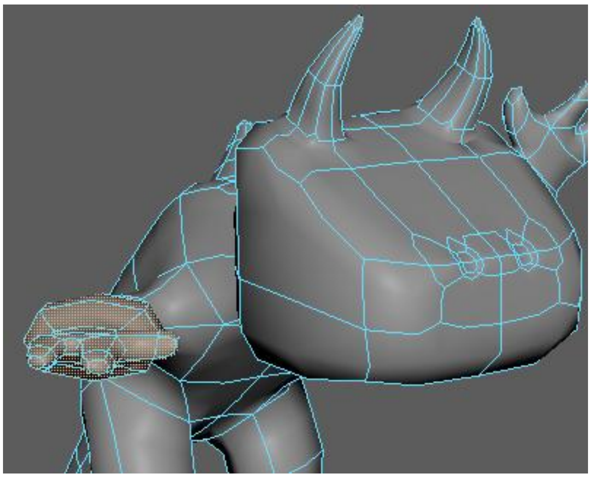




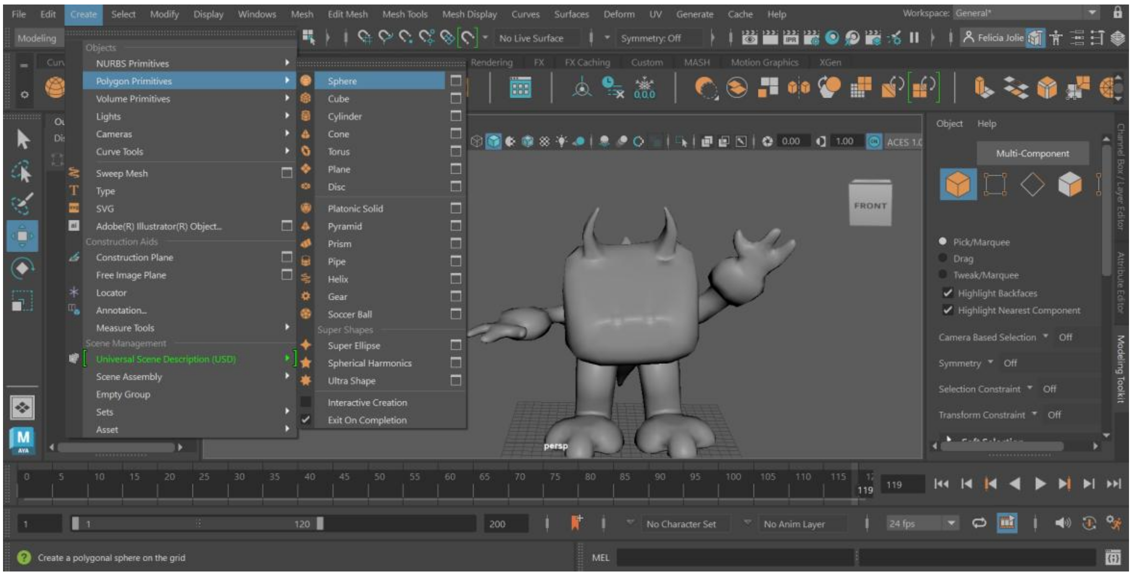
14. I change the placement of the right hand to make it upward by using the move and rotate tool.

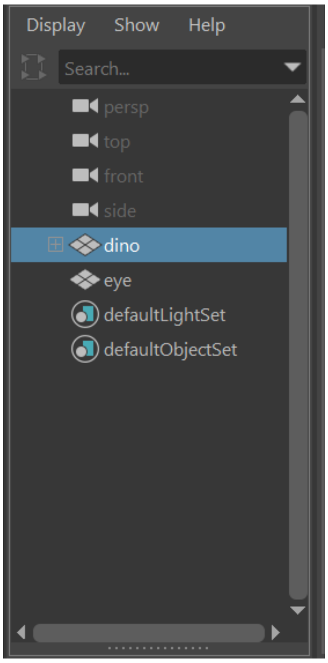
15. I also change the placement of the left hand by making it goes forward.

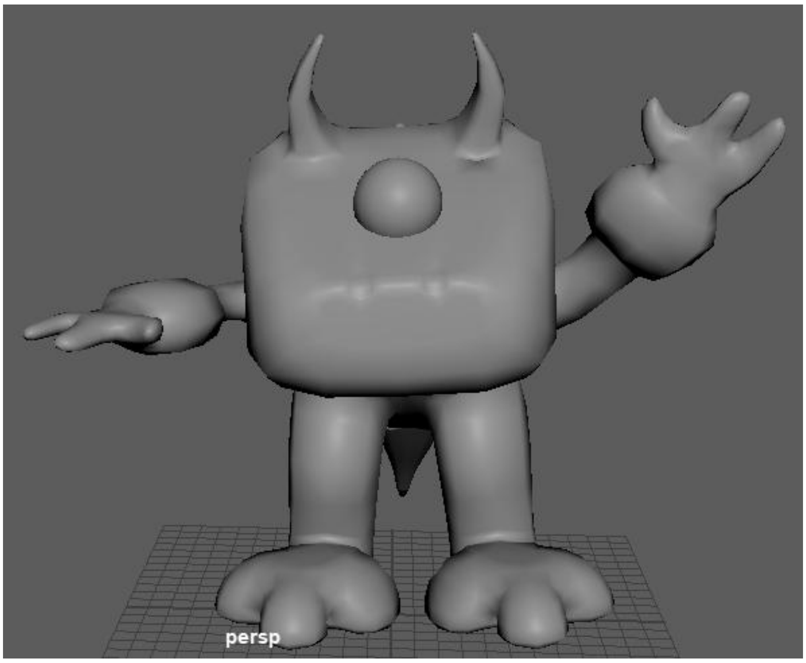



16. I create a sphere by using Create → Polygon Primitives → sphere to make the eye.

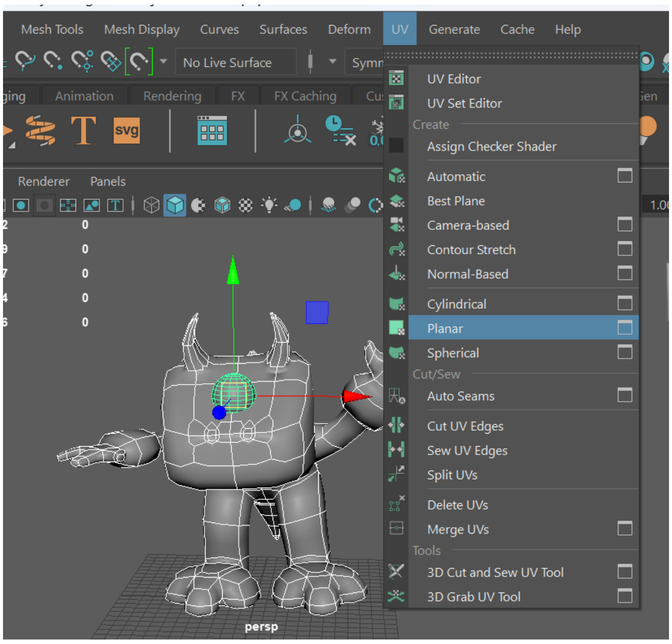


17. I resize and move the sphere using the scale and move tool to place it at the top of the head, so it looks like an eye. I rename the pCube1 to dino and the pSphere1 to eye. Then I smooth my model.

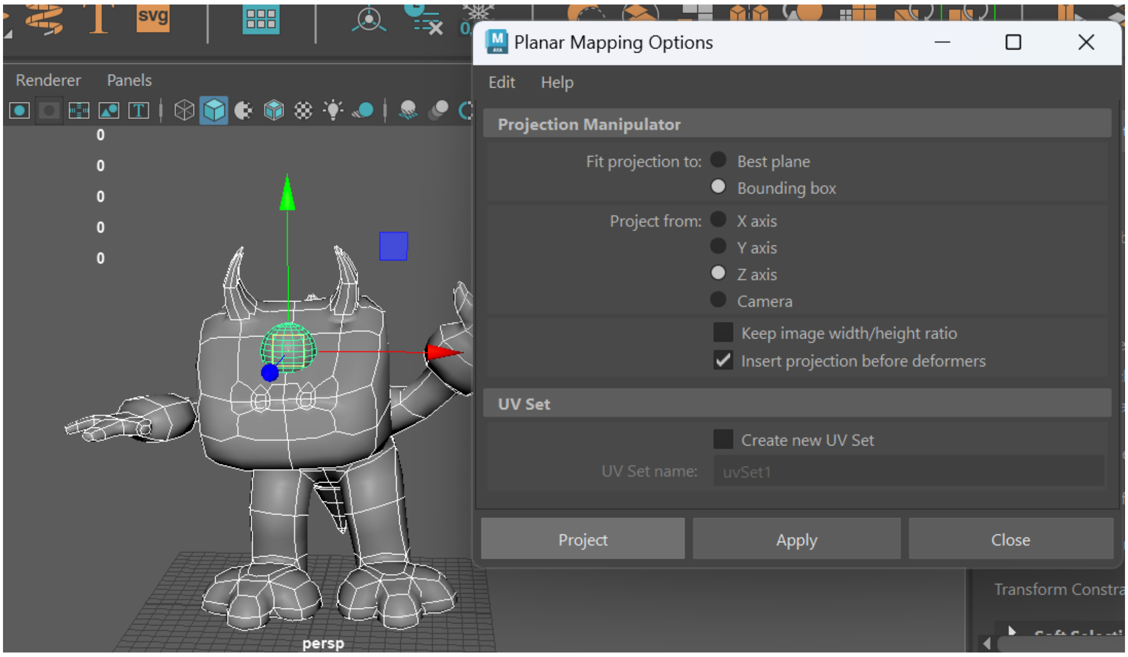




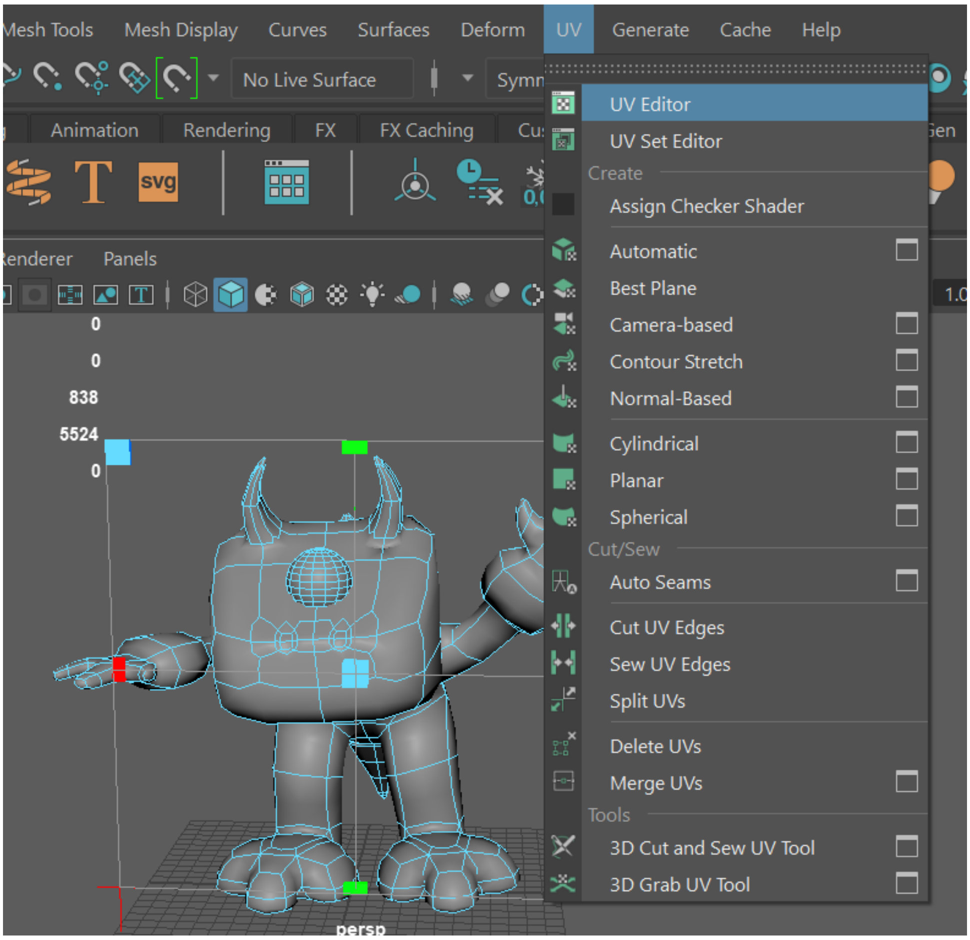
18. For the UV map, I select the project and go to UV → Planar.



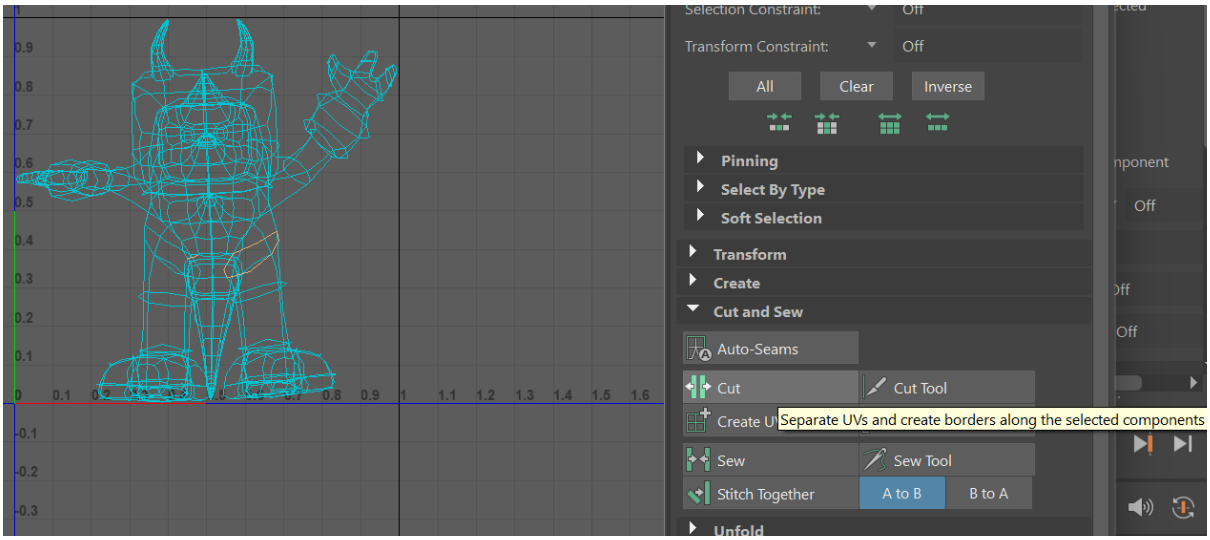
19. Select the object, and then go to UV → Planar → Options, choose the Z axis option, and then press Project.



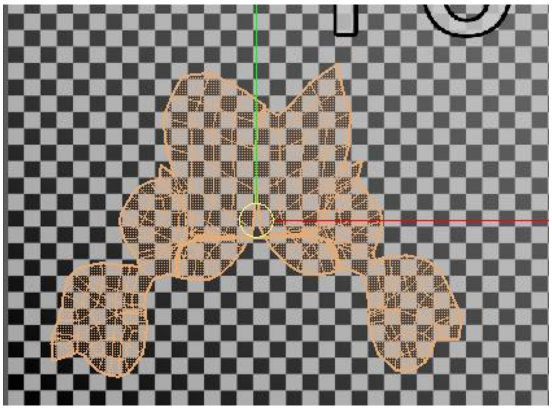
20. I go to UV → UV Editor to start UV mapping.



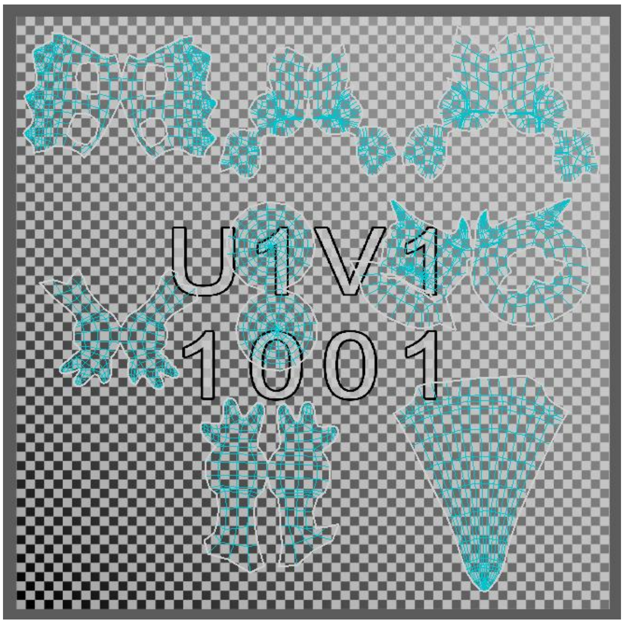
21. I first go around the edge of the leg then I go to UV editor and cut the leg of my model.

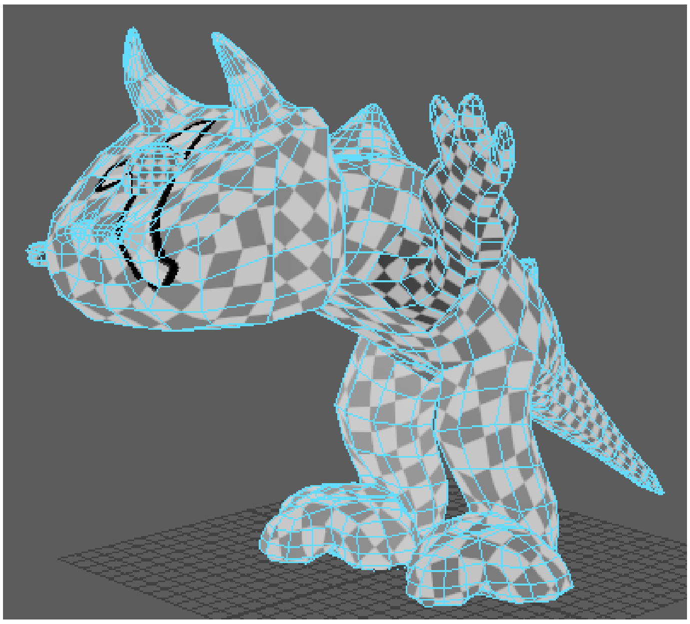


22. Then I cut around the middle of the leg, leaving one edge uncut. This allows me to unfold the piece while keeping it intact as one piece.

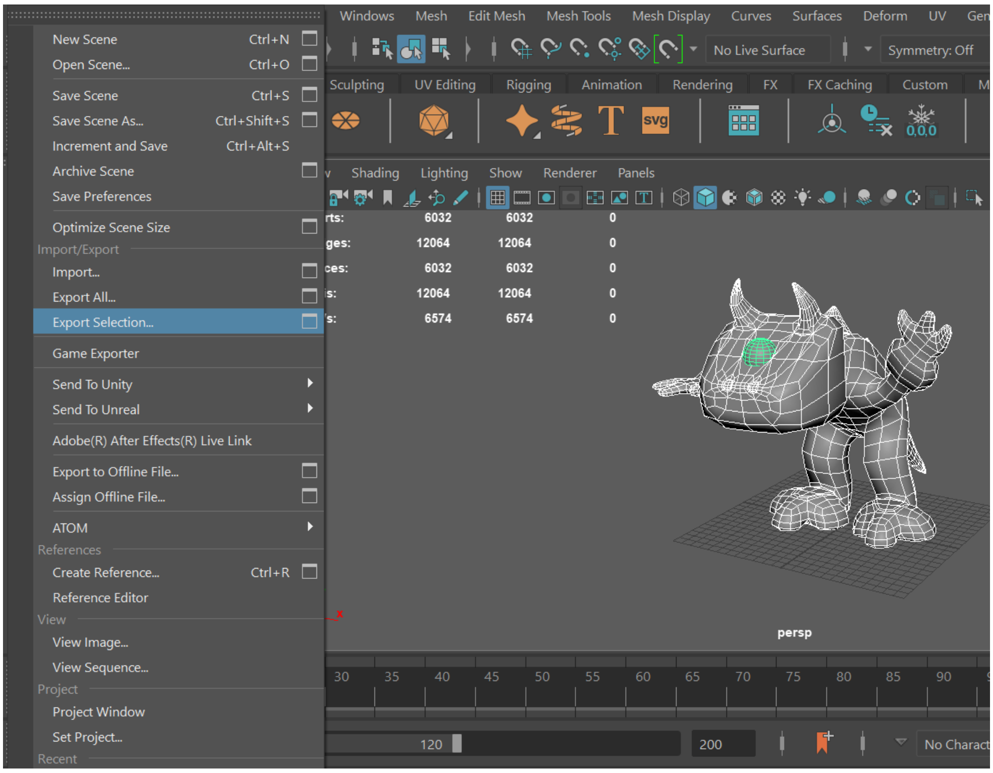


23. I cut and unfold all the other parts of my model until we unfold them properly. We can check if we unfold them properly by toggling the checkerboard pattern on and see that the checkerboard pattern in the model is correct and neat.

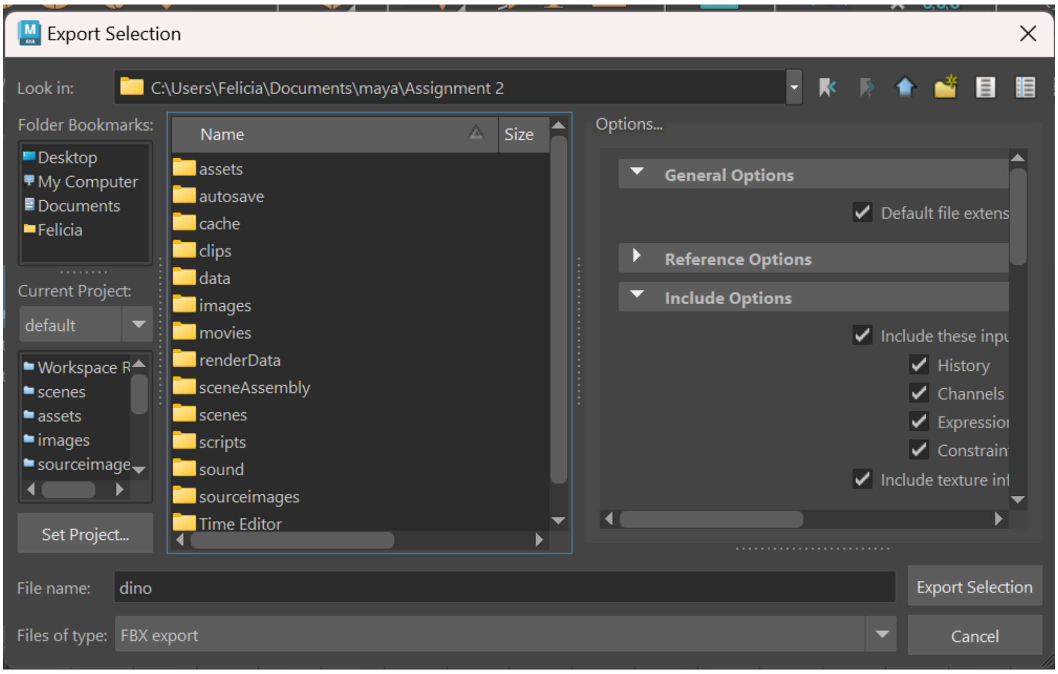




24. Select the object then I go to File → Export Selection to export the object so that I can colour it in Adobe Substance 3D Painter.

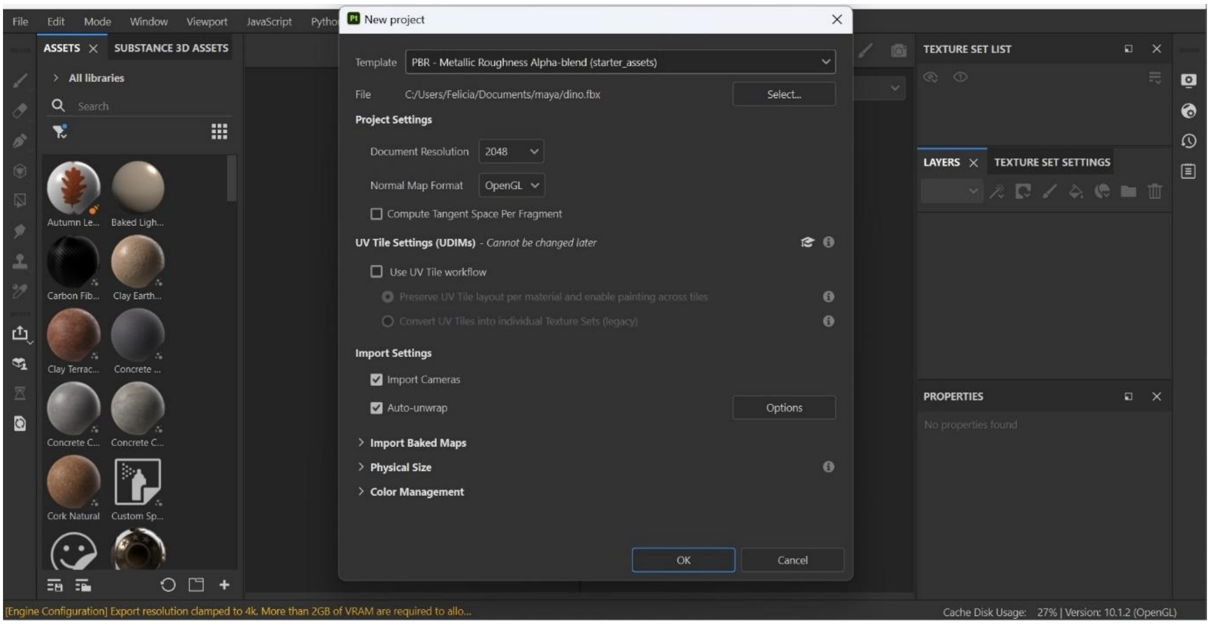


25. I name the file as dino and save it as a FBX export file.

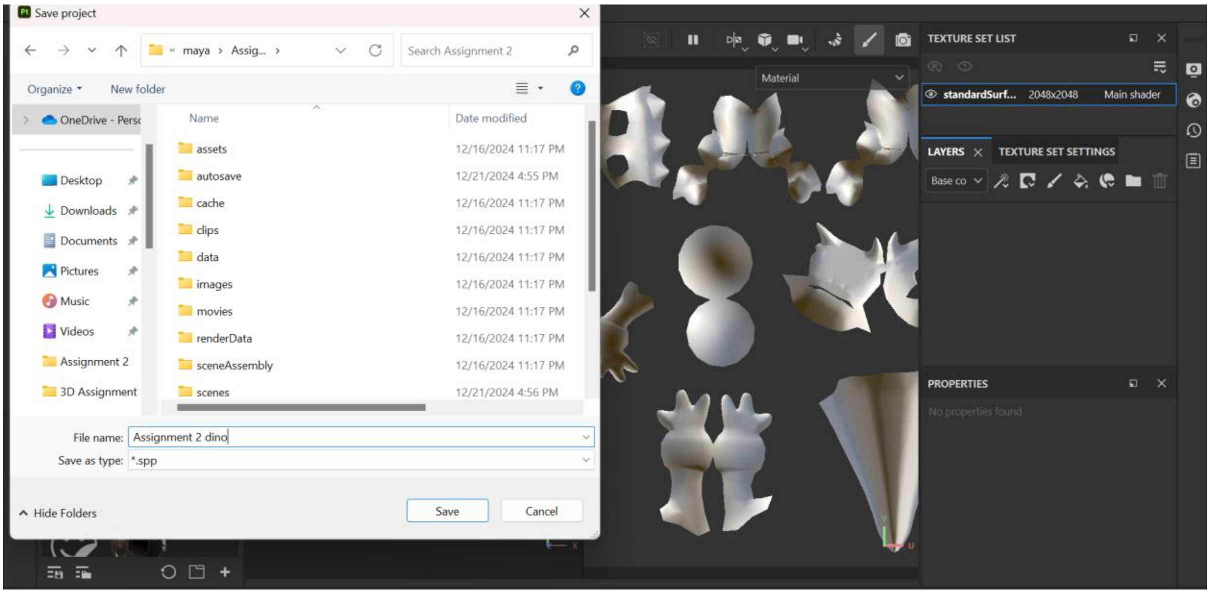


4 Adobe Substance 3D Painter Colouring Process

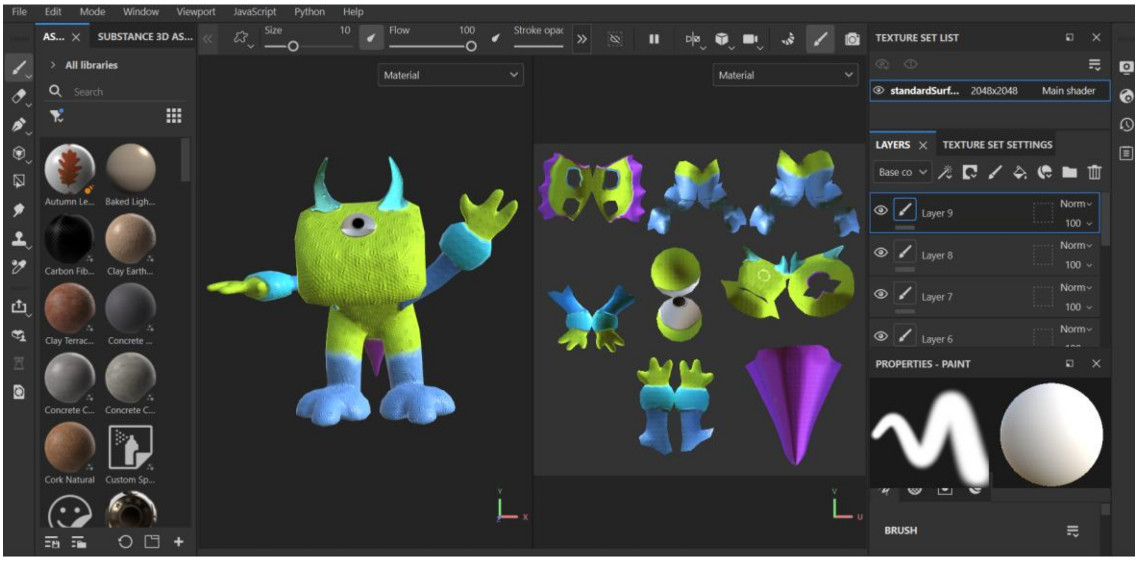
1. First, I create a new project in Adobe Substance 3D Painter.
   Then I select the Template → PBR – Metallic Roughness Alpha-blend (starter_assets), the File to the maya file that I exported (dino.fbx), Document Resolution to 2048 and the Normal Map Format to OpenGL.



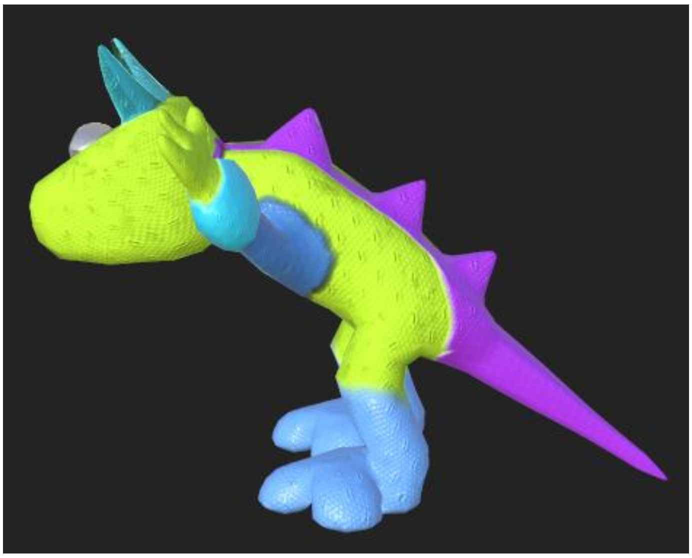
2. Then I save the project as Assignment 2 dino.



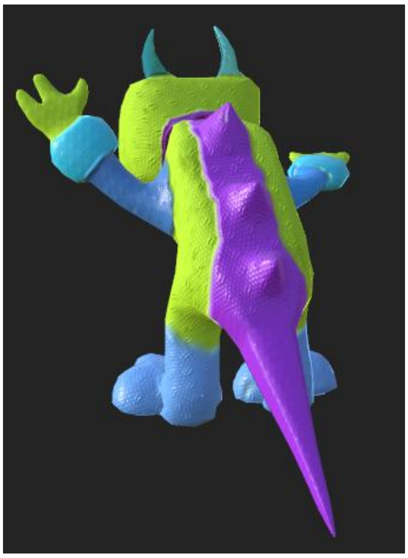
3. I choose the texture for my model and colour it based on my reference which is the aliens from toy story. For the color scheme, I use green as the base color, with light and dark blue for the hands, legs, and horns, and purple for the tail and spike on the back. I also create separate layers for different parts of the model to apply color individually to each section.



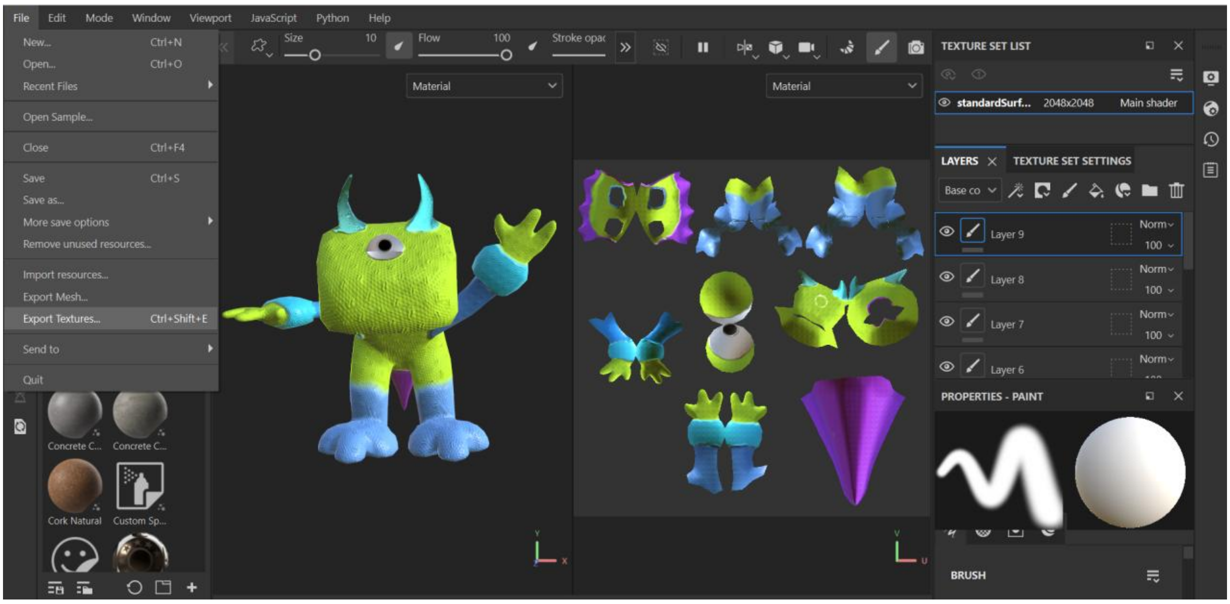
Side View:



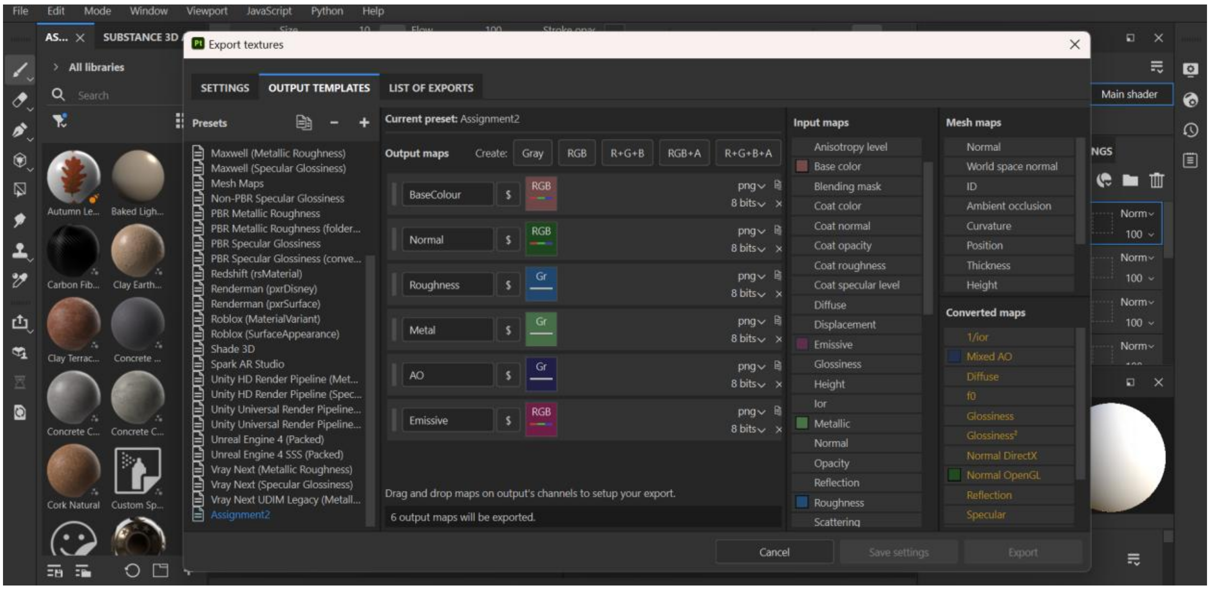
Top View:



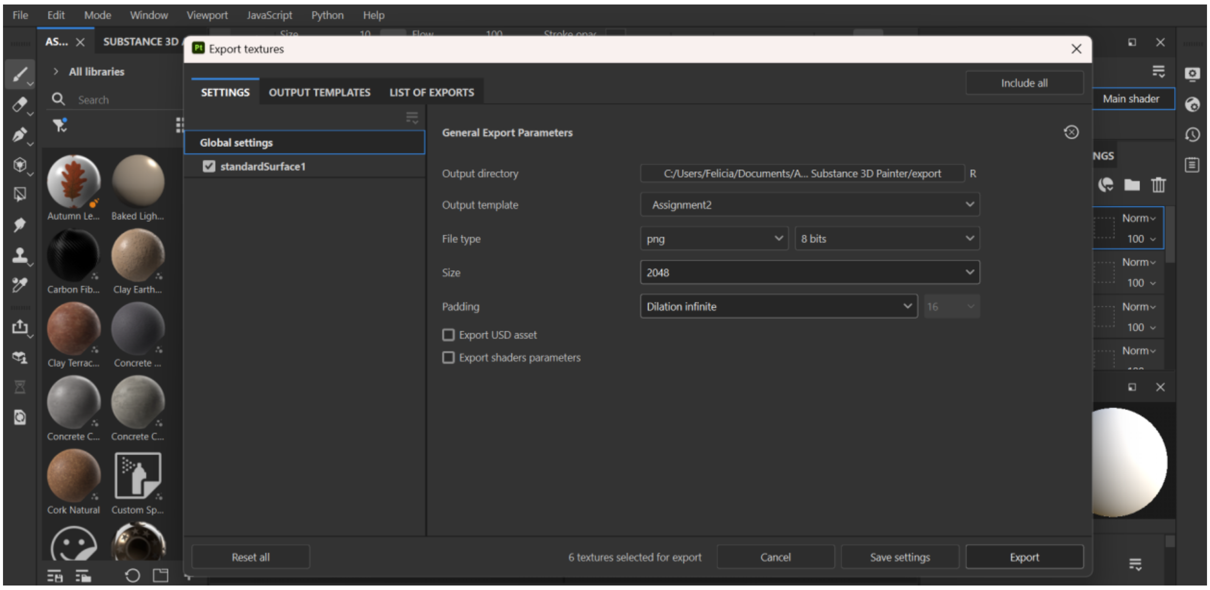
4. Then to export the model I go to File → Export Textures.



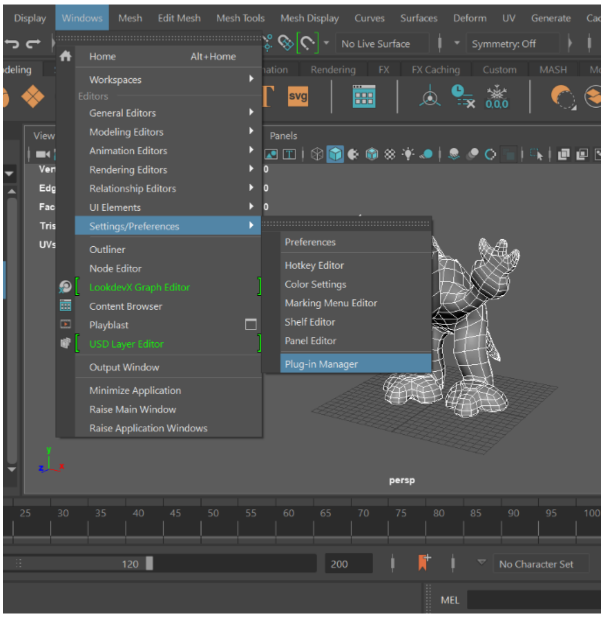
5. I go to the Output Template tab then create a new template and remane it Assignment2. I click the RGB button 3 times and name it BaseColour, Normal and Emissive. I click the Gray Button 3 times and name it Roughness, Metal and AO. Then I click and drag Base Color into the BaseColour RGB square, and select "RGB Channels" from the drop down. I do the same for Normal → OpenGL and Emmisive → Emmisive. For the grey channels, I drag Roughness → Roughness, Metal → Metallic, and AO → Mixed AO.

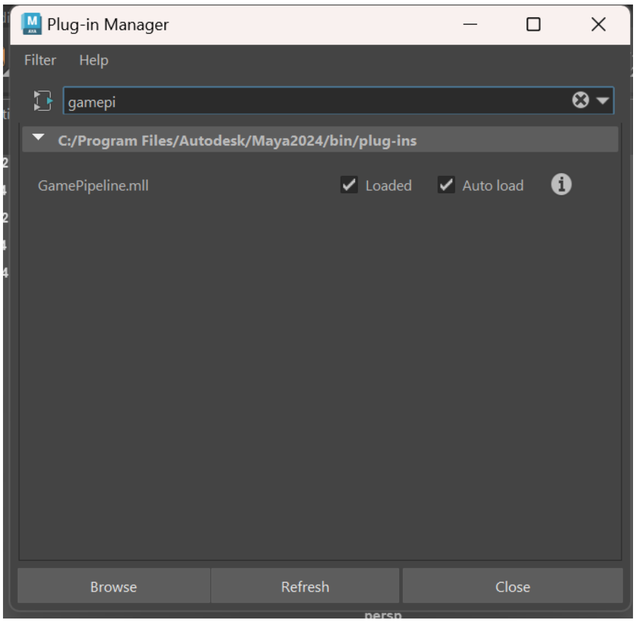


6. I go back to the Settings tab and change the Output Template to Assignment2, the file type to be png and 8 bits, the size to be 2028 and the padding to be Dilation infinite. Then I click export.

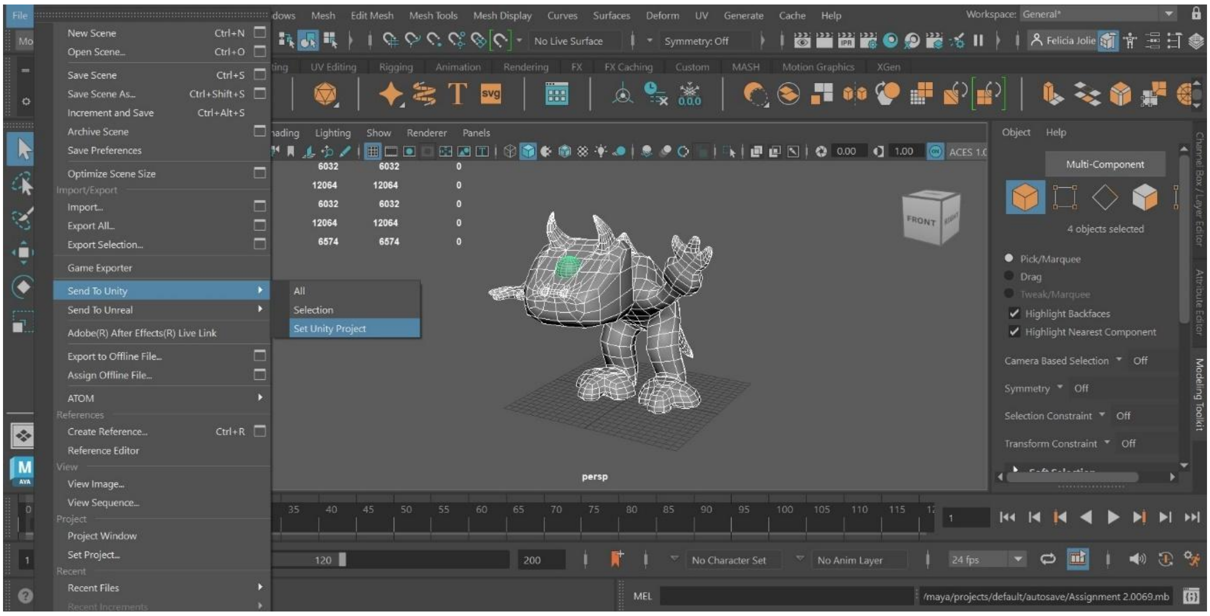


7. Open your project in Maya and ensure that the GamePipeline.mll plugin is loaded. To do that go to Windows → Settings/Preferences → Plug-in Manager then check both Loaded and Auto load.

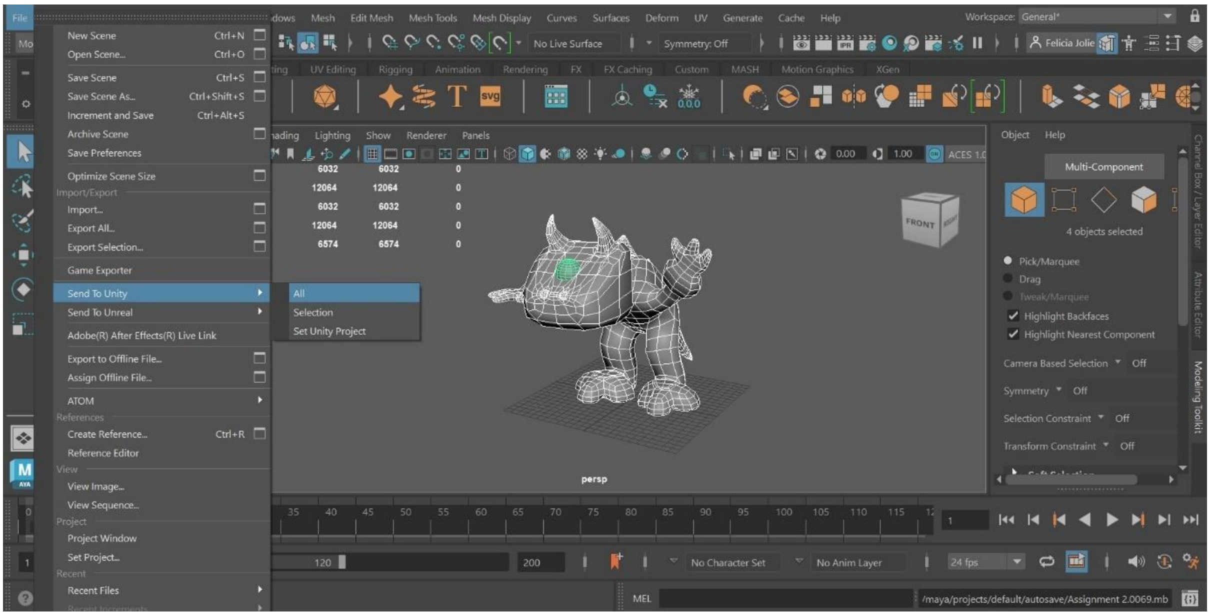




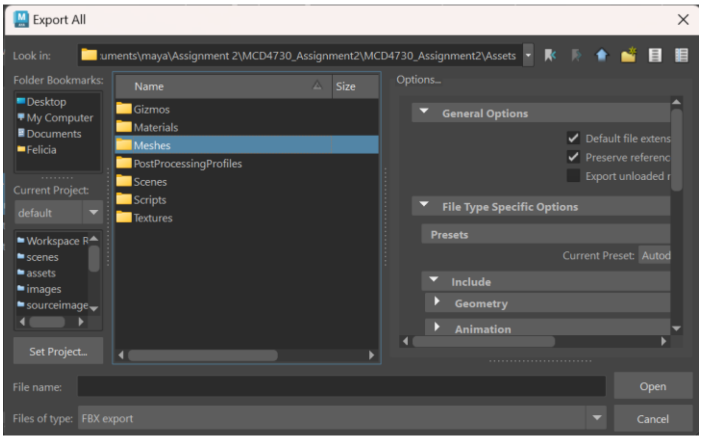
8. Next, we send out Maya project to Unity, first we need to tell Maya where our Unty Project is located. We do this by selecting File → Send To Unity → Set Unity Project then select the same destination where we export our Maya.

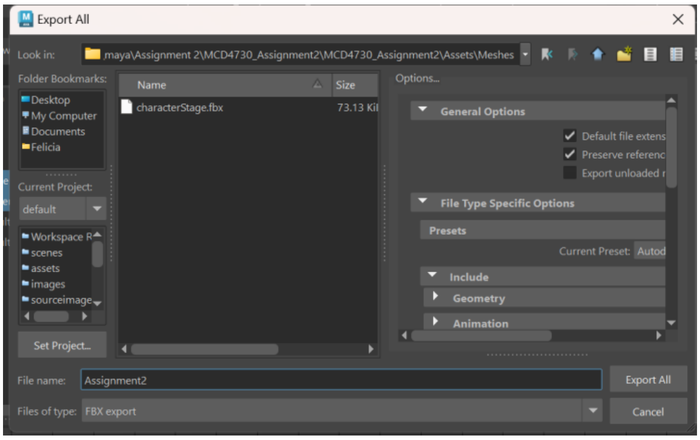


9. Then we send all our mesh to unity by slecting File → Send To Unity → All.



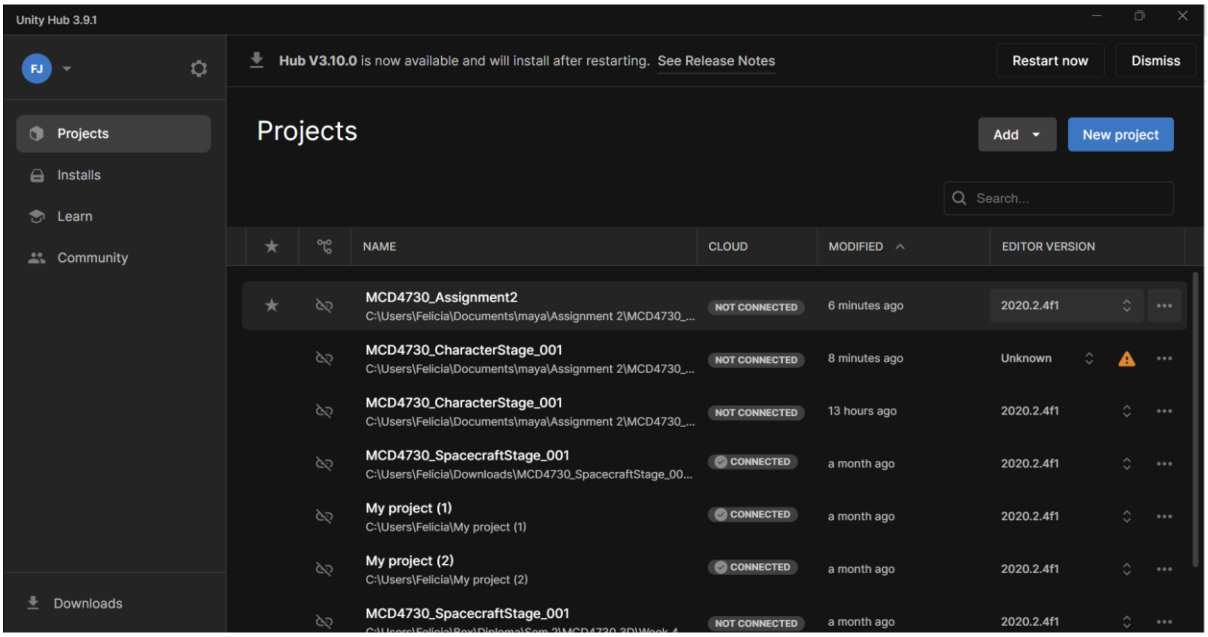
10. We save our mesh in the Assets → Meshes and name it Assignment2.



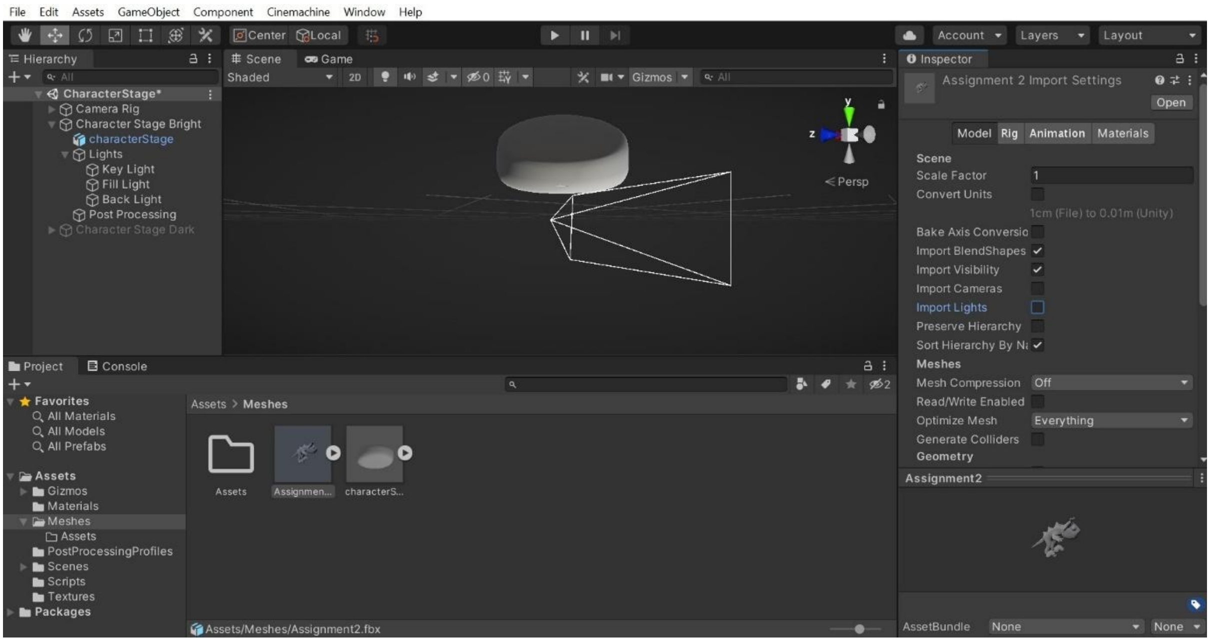


5 Unity

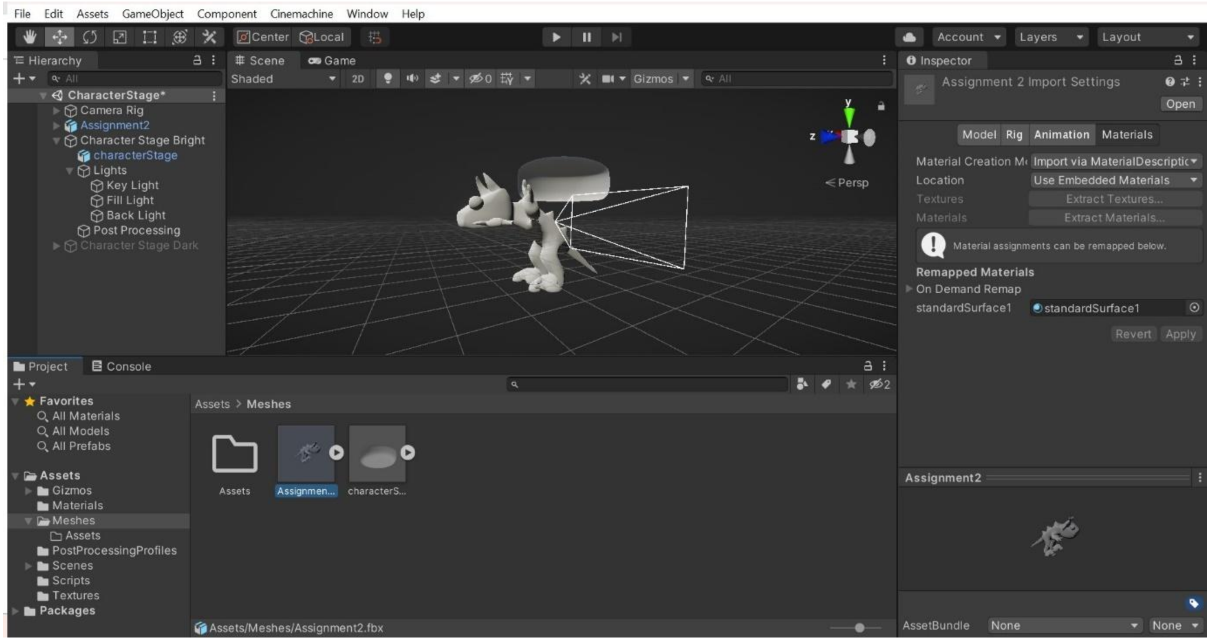
1. Go to Unity Hub and open the MCD4730_Assignment2.



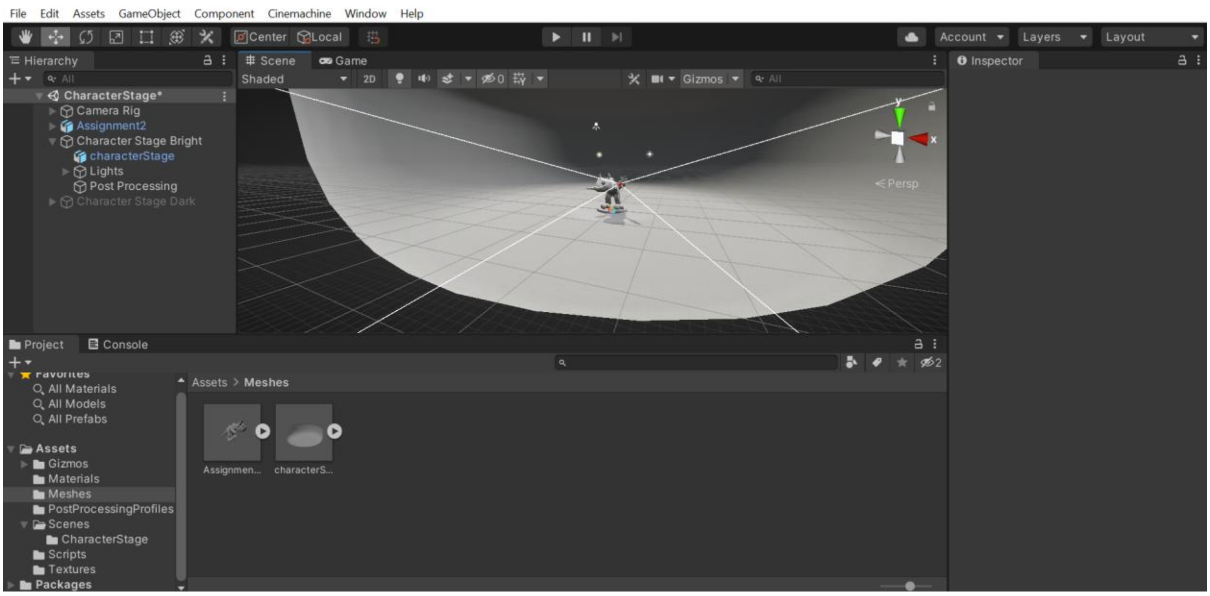
2. In Unity open select the Assignment 2 then go to the Inspector to untick Convert Units, Import Cameras and Import Lights.



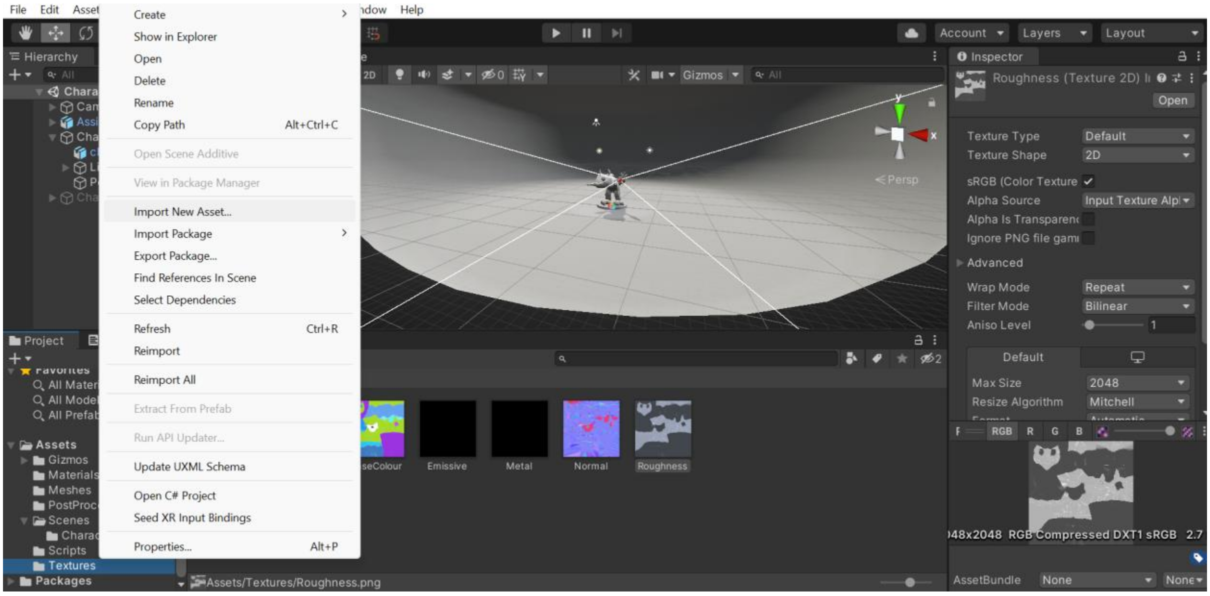
3. In the inspector go to the Materials tab click the Extract Materials. The save the materials in Assets → Materials. Then drag the character form the Project tab and drop it into the scene.



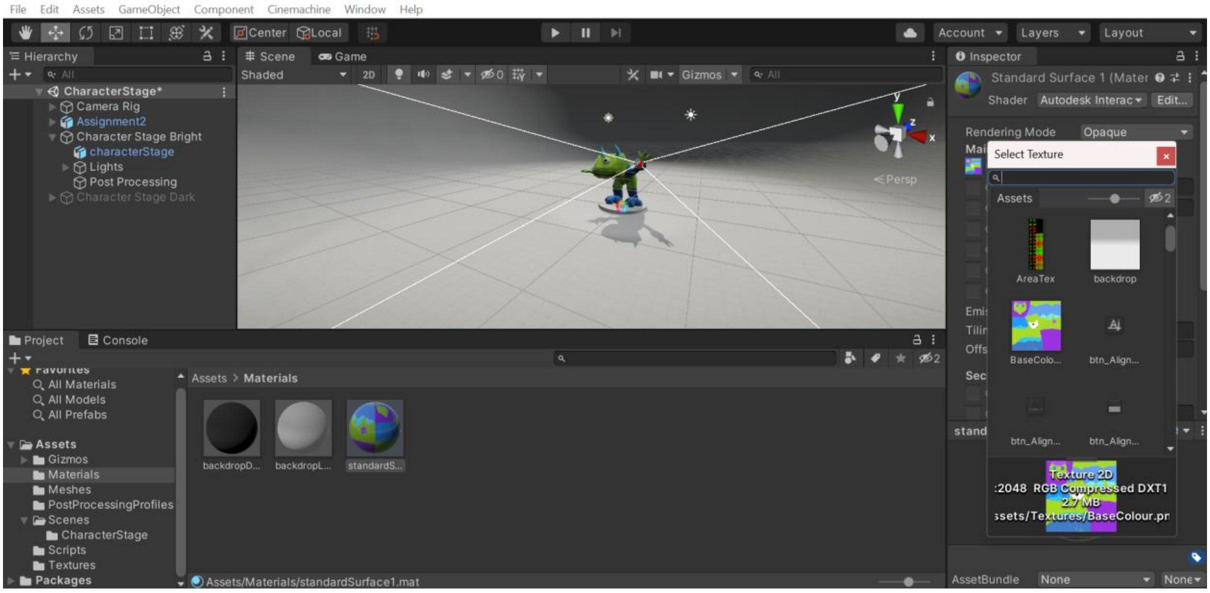
4. Move, rotate and scale the character to position it on the platform in the centre of the world.



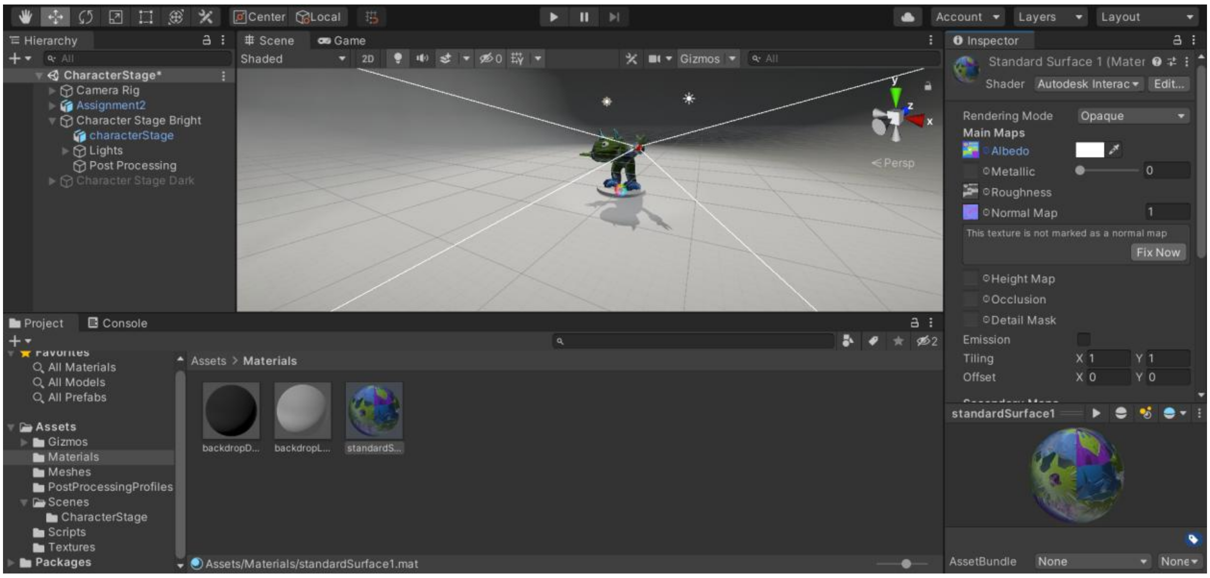
5. Go to Textures in the Project tab then right click to Import New Asset. Import the BaseColour, Emissive, Metal, Normal, Roughness and AO.



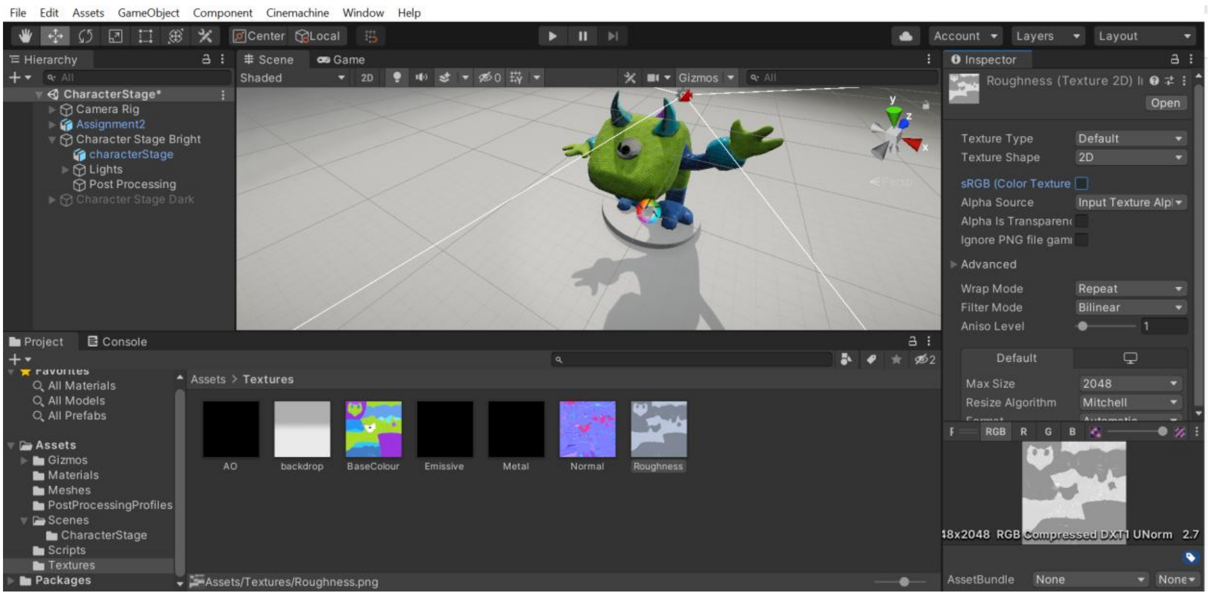
6. Go to Standard Surface 1, which is our character's material and change the Shader to Autodesk Interactive, the Rendering Mode to Opaque and the Albedo colour to White. Then click the small circle next to Albedo and select our Base Colour. Do the same for Roughness and Normal Map.



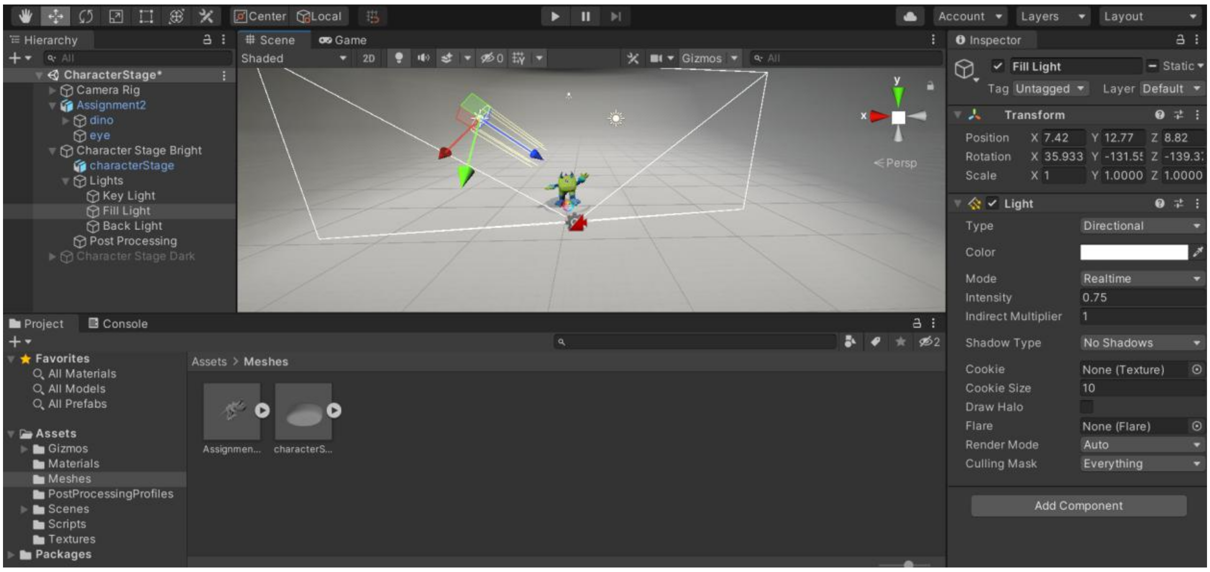
7. When you do that in Normal Map there will be an error that say, “This texture is not market as a normal map”. Click the Fix Now button to fix it.



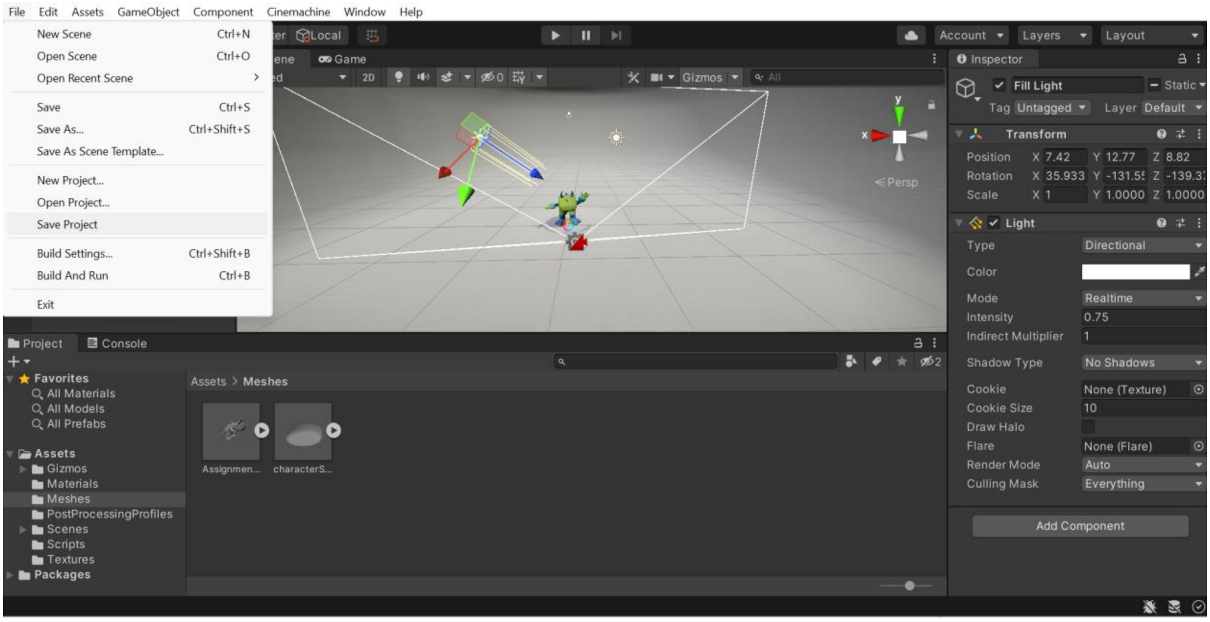
8. Go back to Textures folder and turn the sRGB off in Roughness, Metal and AO texture maps.



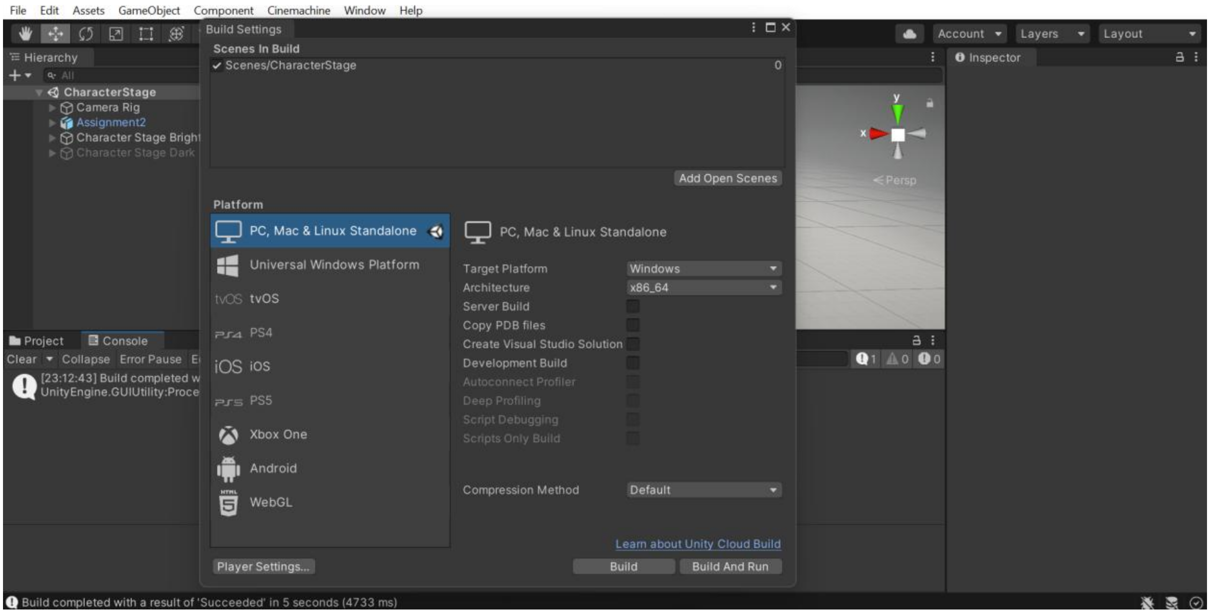
9. I experiment with the lights until I find the lighting that I like.



10. Save the project by going to File → Save Project.



11. Build the project by going to File → Build Settings. Click PC, Mac & Linux Standalone. Make the Target Platform into the intended platform, in my case it is Windows. Then click the Build button.



6 Complete Build

Complete Maya Build for Assignment 2:

Front View:

Right View:

Left View:

Back View: